RT OF INVESTING

BY

A NEW YORK BROKER

LIBRARY
OF THE
UNIVERSITY
CALIFORNIA

NEW YORK
D. APPLETON AND COMPANY
1888

H6452
.119

MC

COPYRIGHT, 1888,
BY D. APPLETON AND COMPANY.

PREFACE.

THE appearance, over his signature, in leading magazines, of the papers that principally make up the following *brochure*, has involved the writer in considerable correspondence. A number of parties have, in consequence, written to him: some requesting a fuller statement of his views on points partially discussed or merely touched upon; some disputing his conclusions or having suggestions of their own to offer; and others, who had been made aware of the existence of the publications through news-

paper notices or in some other way, and not caring to be at the trouble and cost of procuring the periodicals in which they were to be found, asking for a reproduction of such passages as might be applicable to their special circumstances. The interest, as evidenced by the correspondence referred to, which the papers as magazine articles have awakened, seems to the writer to be ample apology for their reappearance in more accessible and economical form if the discussion of a subject of such general concern as the investment of money, with a view to an income more or less fixed and permanent, calls for apology.

Somewhat in the nature of a supplement, has been added a chapter on speculation. People having idle money

often find it difficult to decide whether they had better invest it securely, content with the amount they have and the revenue to be derived as interest, or seek to add to the principal by ventures necessarily more or less uncertain. The temptations to the latter course that are held out by our stock and other exchanges are very dazzling—sometimes quite irresistible. It is to those institutions that persons of a speculative turn and with money in hand are apt to look for the realization of their hopes. To any who may be halting between the policies spoken of, with an inclination to do a little in stocks, the subjoined criticism of the methods and bearings of our leading exchanges that make a business of shares and bonds may prove

to be both entertaining and instructive reading.

A few alterations have been made in the original text, and some additions to cover points raised or suggested by communications from friendly hands; otherwise, the matter is unchanged.

As the author prefers to avoid the suspicion of using this publication as an advertisement of his business—that of a broker in securities—the matter which in another form appeared over his signature is now presented without that appendage.

CONTENTS.

CHAPTER I.

	PAGE
Investing	9
Governments.	43
State Obligations	46
Municipal Obligations	56
Railroad Mortgages	65
Railroad Stocks	72
Other Stocks	74
Farm Mortgages	79
Ranch Securities	92
Water-Works Loans	96
Street-Railway Bonds	107
Miscellaneous Bonds	109
Mining Securities	111
Bridge Bonds	113
Substitution Securities	115

CHAPTER II.

PAGE

SPECULATING 118

APPENDIX.

INVESTMENT SECURITIES 155

NEW YORK STOCK-EXCHANGE . . . 157

PHILADELPHIA STOCK-EXCHANGE . . . 185

BOSTON STOCK-EXCHANGE 190

BALTIMORE STOCK-EXCHANGE . . . 196

THE ART OF INVESTING.

CHAPTER I.

*INVESTING.**

"How can I invest my money to make it pay a fair interest, and at the same time insure its safety?" is a question daily asked by thousands. With the multiplication, consequent upon the growth of wealth among us, of that class of persons who want to live by their means, without care or labor, the number of anxious inquirers on that point is

* The greater part of this chapter appeared in "The Popular Science Monthly" under the title of "The Art of Investing."

constantly increasing. It would seem, when reference is had to the many securities, both bonds and shares, that are offered, often at temptingly low prices, to be a question very easily answered. The truth is, that there is none more difficult. The ordinary investor who goes about the work of converting his cash into paper combining the two elements of value spoken of, finding himself hopelessly embarrassed by the seeming richness of the market, soon gives up in despair, and turns the job over to some banker or broker who works for a commission. Experience shows that even then he is too often the victim of defective judgment or misplaced confidence.

If it is difficult to make money—a proposition about which there will not be

much diversity of opinion—it is in most cases even more difficult to keep it profitably employed. Men of prudence and skill in the acquisition of capital often show astonishing recklessness in the disposition they make of it. The strangest caprices take possession of them when it comes to the critical moment that calls for a choice of investments. And as riches are always clothed with folded wings ready to expand at the most unlooked-for exigencies, it is not much wonder that they frequently take to the winds and pass beyond recall.

The history of investment securities would furnish a most interesting study. In no other department of business have there been greater changes. The time has been, within the memory of many

now living, when the man who had money to put at usury generally loaned on personal indorsement, the borrower relying on his neighbor or other good friend to "back" his paper for him. The mortgage on real estate, of course, was known; but, owing to the short intervals for which loans were generally made, was not often resorted to. The shares of banking, turnpike, canal, railway, and other incorporated companies after a while began to absorb the money of people who wanted to realize more than current rates of interest, and were willing to take corresponding risks.

Nor have we yet reached a finality in the matter of investments. Changes and innovations are of continual occurrence. Not only are new securities coming upon

the market, but new subjects as a basis for their production are industriously sought after, and new forms for their preparation are being invented. Such things as commercial farm mortgages, water-works loans, street-railway debentures, etc., etc., were utterly unknown but a short time ago. Progress in the manufacture of investment issues keeps pace with all material developments. Every newly-discovered force or process in mechanics means the appearance of another detachment of paper securities.

The war of the rebellion popularized the coupon bond, in consequence of its adoption by the Government, and made it the favorite form of investment paper. Railroad and other corporations lost no time in availing themselves of the confi-

dence which that species of debenture inspired, and States, cities, counties, etc., were soon flooding the country with obligations carrying long coupon attachments. Except for Government and municipal uses, there never was a more disastrous invention. It has been the means of numberless deceptions, and has inflicted heavier losses upon the investing public than all other devices combined. Being supplemental to stock certificates, it has duplicated representatives of the same values and led to excessive issues of paper; it has separated capitalists from the management of properties into which their moneys have gone; and, being based upon mortgages promising absolute security, it has too often accomplished the grossest decep-

tion. Many a man has purchased and paid a good price for a mortgage coupon bond, giving him no control over his security, who would have rejected a share-certificate standing for an equal interest in the property pledged, and giving him the right to participate in its management, with the possibility of a greater return for his money.

Under the careless legislation of many of the States, which has permitted corporations to decide for themselves the amounts of obligations they might put out, it is no wonder that the privilege has been abused, and the making of shares and bonds, the latter represented to be amply secured by mortgage liens, has been carried to criminal excess. One illustration will suffice:

The Arkansas Central Railway Company (the name indicates the locality) built only forty-eight miles of its projected road. The road was of narrow gauge, with very light iron, and in every way cheaply constructed. It cost less than ten thousand dollars per mile, including equipment. As has been the case with most of companies building railways in new countries, help in its behalf was asked from the communities to be benefited, and their bonds amounting to nearly half a million dollars were given it by counties, cities, etc. Under a statute providing for aid to railroads when their beds could be utilized for levee purposes, the company got $160,000 of State bonds. Under another statute it got, as a loan from the

State, its bonds to the amount of $1,350,000, which were to be a first lien upon the property. After such abundant assistance, it would have appeared hardly necessary for the company to put out obligations of its own. However, it proceeded to issue and market its own debentures to the amount of $2,500,000, of which $1,200,000 purported to be secured by first mortgage, a representation that, for reasons already stated, was not correct. In addition, a considerable amount of stock certificates was issued. Altogether, nearly $5,000,000 of paper was put out and negotiated on the basis of forty-eight miles of narrow-gauge road. But this proved to be insufficient. The road, for non-payment of interest on its bonds, soon passed into the hands of

a receiver, who found it in such an unfinished state that, with the court's permission, he issued a considerable amount of his own certificates to provide for necessary repairs and betterments. Then the road—the product of so much outlay—was sold at public auction, and brought the magnificent sum of $40,000, which was paid, not in cash, but in receiver's certificates that had been purchased at a great discount from their face!

The business of manufacturing securities has not been confined to railroad builders. We now have stocks and bonds upon the market representing nearly all conceivable kinds of property—telegraphs, telephones, mines, cattle ranches, grain and grass farms, water-

works, bridges, oil and gas wells, electric lights, factories and mills of every description, patent rights of all sorts, steamboat lines, apartment houses, and, in one instance, a cemetery! And not only are properties of many kinds used to issue bonds upon, but many kinds of bonds are often issued upon the same properties. Thus we find among our railroads not only first, second, and third mortgages, but income bonds, convertible bonds, consolidated bonds, redemption bonds, renewal bonds, terminal bonds, divisional bonds, sinking-fund bonds, "blanket-mortgage" bonds, collateral trust bonds, equipment bonds, and bonds *ad nauseam*, until they lap and overlap in seemingly endless complication. Not that merely, but one issue

of bonds is sometimes made the basis for other issues. Indeed, one of the money-making devices of the time is the formation of companies that issue their bonds on the security of other people's bonds that they have purchased, either yielding a higher rate of interest or obtained at lower prices than they expect to realize for their issues. There seems, in fact, to be no limit to the production of securities that are spread before capitalists. There never was a time when it was so easy to invest money — and to lose it. Of the securities that are offered with first-class recommendations, it is probable that about one third are actually good, one third have some value, and one third are practically worthless. Hence the very natural inference that

whatever art there may be in the matter of investing is to be exercised chiefly in the avoidance of unworthy offerings, and it is to that point that a profitable discussion of the subject must be mainly directed.

For the condition of things described, the laws of some of our States in giving corporations almost limitless power to issue negotiable paper, as well as in permitting all sorts of companies to incorporate themselves, are, undoubtedly, very largely to blame. Our banks are closely watched and very properly restrained from taking people's money on false pretenses; but how much better is it for railway and other corporations to take it by means of legalized fictitious evidences of value? Banks

are by no means the only corporate institutions that need watching. One of the reforms that would seem to be very much demanded is legislation prohibitory of the creation by companies existing by authority of law, of debentures or scrip not representing moneys actually paid into their treasuries, or proprietary interests whose values are to be determined by disinterested parties. Pennsylvania has incorporated substantially such a provision in her constitution. Her example should be followed by all other commonwealths. But as long as corporations are permitted, almost without limitation, to flood the country with seductive promises of money payments and to have them offered in all our financial marts, it behooves

investors to be correspondingly cautious in making their purchases. If the law will not protect them, they should be the more careful to protect themselves.

But the security behind or beneath the debenture or other paper obligatory is not the only thing to be looked into by the investor. Even the form of the document may be important. A case in point, inasmuch as it shows how the preparation of an undertaking for the payment of money may change its apparent value, would seem in this connection to be appropriately quoted. Some years ago certain townships in the State of Missouri were desirous of aiding the construction of railroads with their credit. The State legislature, to that end, passed an act authorizing the

issue and sale of bonds obligatory upon them; but it was stipulated—a very singular provision—that, instead of being put out by the townships, the bonds should be executed by the officials of the counties in which they were located. Accordingly, debentures aggregating several million dollars were thus prepared and disposed of. The bonds bore the seals of the counties and the signatures of their officials. On the back and at the top of each obligation in large letters were the words "county bond." The instrument began with the recital, in the usual form, that it was issued by the county; but farther on, and in the smallest type employed, came the statement that it was executed for and in behalf of a certain township, which alone

was to be responsible for its payment. These bonds were extensively advertised as "county bonds," and probably in most instances, certainly in many, were sold as such, and it was not until purchasers had parted with their money that they discovered that, instead of getting the paper of well known and wealthy counties, they had secured only the obligations of townships they had never heard of before. It was then manifest enough that they had been made the victims of a piece of very sharp and very shabby practice. In very many cases the buyers of bonds and other securities learn, when it is too late, that their purchases; owing to some obscure and apparently innocent passage that had been overlooked or

disregarded, are very different from what they thought they were getting. How often have careless investors that supposed they were purchasing undertakings that would be good for long terms of years, and probably paid premiums to obtain them, ascertained at the end of comparatively short intervals that they were forced to accept in payment the amounts nominated in the bonds in consequence of unnoticed clauses giving their makers power to redeem at their option? The lesson of such cases is obvious enough. It is that no one should ever buy a moneyed undertaking without having first carefully read it. This may seem like an unnecessary warning; but in truth it is a most material one. Thousands and

thousands of dollars have been lost by the neglect of this simple precaution. "I didn't read the paper" is the explanation that has again and again been offered when time has disclosed a different investment from the one intended to be paid for. The fact is that comparatively few unprofessional bond and share purchasers ever carefully examine the instruments they acquire. They look at the headings, those parts that are in big letters, and take the rest for granted. It is a most unwise practice. Unless you are previously familiar with the document in all its parts, don't fail to read it before you buy. Read it all, the little type as well as the big type, the indorsements, the coupons, and all. Don't take somebody's else word for it.

Examine the seal, the signatures, and even the embellishments. Something may be disclosed that will change your mind and save your money.

But if there are tricks in the making of securities, even more are to be apprehended in the selling of them, and should be guarded against with corresponding diligence. It is a notable fact that no poor securities are ever offered. They are always good so long as they are on the market. It is only after they have been purchased that they prove to be worthless. Interest has never been known to fail on bonds that were seeking investors, although default has sometimes followed very closely on the sale of the last obligation. Indeed, it is no secret that interest is

often paid out of the proceeds of bonds, particularly railroad issues, the purchasers in that way getting a portion of their own money back while the process of marketing them is going forward, although such a thing has seldom been known to happen when the entire issue was disposed of. The advertisements of the bond-sellers are sometimes marvelous productions. No such securities as they have to offer have ever been on the market before. They are absolutely safe; they pay extra rates of interest, etc., etc. The wonder is that, with so much capital seeking investment, it is found necessary to advertise such perfections at all. In such cases it is hardly necessary to say that the only safe rule for investors is to find

other uses for their money, however strong the temptation may be.

A common expedient of bond-makers and bond-merchants is to fortify their issues with the favorable opinions of eminent lawyers. This is particularly the case when the obligations of municipalities, or of companies that are dependent upon contracts with municipalities, are offered, municipalities having shown an unpleasant disposition to go back on their undertakings. No exception can be taken to the practice referred to, as counsel learned in the law should in such cases always be consulted; but the writer has to say that he has never yet known a security so poor that a lawyer's opinion could not be had to back it. Such testimonials should be

taken for what they are worth, and no more.

When so many seductive baits are offered; so many nets and traps, contrived and constructed by clever brains and cunning fingers, are spread for the capture of those having money, is it surprising that the careless and credulous are victimized, and even that the sagacious and prudent should sometimes be taken in? Nevertheless, for the losses they have sustained, investors, as a rule, have themselves chiefly to blame. The mistake made, in nine cases out of ten, has been the purchase of *cheap securities.* The hope of realizing a little more than ordinary interest, by buying paper at a discount, has proved to be the rock on which unnumbered capitalists have split.

In addition to their money's worth, they have endeavored to get something for nothing, with the result most generally of getting nothing for something. It is remarkable how blind are people, ordinarily sagacious enough to make money, to the fact that property can not pay a revenue beyond its producing capacity. For instance, how can a railroad company, whose line is wholly or mainly built from the proceeds of mortgage bonds, sell them at a heavy discount, besides allowing large commissions for the selling, and then pay a high rate of interest on their face? Or how can a poor agriculturist, occupying a half improved farm out on the frontier, with a family to support, and grain selling barely above the cost of production, pay

ten or twelve per cent. upon the capital with which he does business.

By what rule or rules is the investor to govern himself? No formula can guarantee him absolute safety. One thing, however, he can properly count upon, viz., that he must expect to pay a fair price for a good security—one that will return him no more than a moderate interest on his money. If he wants to speculate, and is willing to take risks, that is another thing. He can then look for bargains. As a general proposition, it can be asserted that the day for high prices for money, as the day for large profits in trade or manufacturing, or, indeed, in any regularly recognized business with us, has gone by. The capitalist who sends his

money into a new section, or puts it into a new mechanical process or a new constructive enterprise, may or may not make a hit, but for the ordinary and conservative operator the condition of the commercial and financial world gives warning that only small profits are to be looked for. The first and main thing to be studied is safety. And yet there is such a thing as going too far in the matter of prudence. The investor may pay too dearly for safety. There are securities which, compared with others that are to be had, sell at prices much above their real value. The reason is that everybody knows them to be good, and investors who don't want to take the trouble to investigate, or are afraid to trust both their own judg-

ments and the counsels of their friends, are willing to pay extra prices for them. But there are plenty of others that may be had at lower figures, which are just as good. There is no reason in the world why the investor should not get at par all the paper he wants that will yield him six per cent. interest, and be as safe as any property can be under human supervision. As heretofore stated, with the creation of new enterprises and properties, and the development of old ones, new securities are constantly appearing in this country, and a fair share of them ought to be good. Indeed, our securities ought to be the best in the world. The sure and rapid growth of our resources supplies a reliable support, as long as fair intelligence and common

honesty attend their production. The only thing is to choose with discretion, so many doubtful and even fraudulent issues appearing at the same time; but no more judgment is really demanded than in purchasing lands and cattle.

Two common and often fatal mistakes should be avoided. One is in relying solely upon the advice of a broker. No one competent to form an opinion for himself should put his pecuniary interests unreservedly in the keeping of another. Such absolute confidence invites betrayal. By far the greater number of losses to investors has been in securities purchased exclusively on the recommendation of interested commission men. While it is well to get the opinion of a reputable broker, the pur-

chaser should investigate and decide for himself. The other mistake is in giving a preference to "listed" securities. Many persons seem to think that stocks and bonds must have a value if they are quoted at some stock exchange, forgetting how many "fancies" have been ballooned until they have burst at such places. On the contrary, such a position is likely to expose them to manipulation for purely speculative purposes. Stock-exchange quotations, as a rule, are unsafe guides to buyers. They represent not so much the value of the property as the pitch of speculation at the time. When securities are converted into foot-balls for gamblers to play with, they are pretty certain to be either too high or too low. The only ad-

vantage they can have is a readier marketability in case of an urgent need to sell; but it is at the times when such need is likely to exist that they are pretty certain to be at the lowest point. No speculative help can long take the place of real value. Securities, in the long run, must stand upon their merits, and purchasers have merely to follow business principles as taught by the canons of common sense.

In seeking investments, and especially long-time investments, there are several things to be taken into account. There is not only the question of the kind of security to purchase, but the question of the time to purchase. There are opportunities to be looked for as well as pitfalls to be shunned. It is

during periods and seasons of depression, when securities are forced upon the market, often to be sacrificed—and they are certain to come if waited for long enough—that the shrewd investor finds his richest harvest. That, however, can not be said of the ordinary investor. He usually buys when securities are up and confidence is unimpaired, and becoming frightened as market values go down sells when they are at the bottom, and holds his money to reinvest in something else no better, and probably not as good, when the tide has turned. As a rule, the best time to invest is when others are unloading. In money matters it is never safe to follow "the crowd." Nor is it safe (which, however, is little more than the expression of the same idea in

another form) to purchase a security when it is on the "boom." A peculiarity of our money market, conservative as it is popularly supposed to be, is that it is constantly changing its favorites. Its offerings come in waves. Its dealings at one time may be chiefly in railways, at another in municipal obligations, and at another the excitement may run to mining shares or mortgages on ranches and real estate. For the time all professional brokers and bond and share sellers urge their customers to adopt the popular issue, of which, as the result of the increased demand, there is almost certain to be an excessive if not fraudulent production. To yield to the pressure at such a time is always risky. Old and tried securities, like old

friends, are likely to be the truest and best.

One thing the investor would do well never to forget, viz., that there is always plenty of good securities in the market. No one with money need ever fear that others will get all the solid investments, and, in the apprehension that there will not be enough of that sort to go round, put up with an inferior article. Don't let him choose what is not altogether satisfactory, under the impression that nothing else as good or better will offer. If he does so, sooner or later he will regret it. Something good always comes to him who waits with money in his hand.

Another thing of a precautionary nature it is well enough for the investor to

do, and that is to scatter his purchases. The old adage about putting all the eggs in one basket applies with peculiar force to investments. The tendency with those having but moderate sums to place at usury, and who need to be the most circumspect, is to make up their minds in favor of a single line of securities and put everything there. Of course, a failure in that quarter is particularly disastrous. The writer knew a party, some years ago, who decided in favor of municipal obligations, saying that he had satisfied himself that, on the whole, there was nothing else so reliable. Accordingly he put his entire available means into them. But practicing abundant precaution, as he supposed, he divided his money equally among municipal is-

sues of Illinois, Missouri, and Kansas, they having the most paper at that time on the market. He thought he was certainly safe as to part. But soon afterward a wave of repudiation sentiment swept over that part of the country, and every one of his bonds was left in default. It is well enough to scatter in kind as well as in locality.

With these preliminary observations, as a further aid in making a choice it may be well to take up the different kinds of investments and consider them separately.

GOVERNMENTS.

Of these there were at one time outstanding over two and a half billion dollars; and, of course, a great deal of in-

vestors' capital found accommodation there, with satisfactory returns; but the amount has so diminished, and is so rapidly diminishing, while prices have correspondingly advanced, that private parties who want to get the most they can for their money no longer look to that quarter. It is only when a place is sought for trust and other funds for which safety and facility of conversion are wanted rather than a high rate of interest, that governments are to be considered. They are absolutely good and always marketable, but they are no longer to be classed with investments whose claims call for serious discussion.

The foregoing remarks, it is hardly necessary to add, refer exclusively to the obligations of our own Government. If

we pass beyond the limits of this country, we find not only an abundance of government issues that are seeking buyers in the markets of the world, but a corresponding variety in values and quotations. It is estimated that the civilized nations of the globe now owe, without including local or divisional undertakings, not less than $27,000,000,000, a sum which, if equally distributed, would impose an incumbrance of $720 on every square mile of their territory, and a per capita indebtedness of $23 on all their subjects. The foregoing amount has necessarily absorbed a very large proportion of the world's investment capital, and, as its tendency in many localities is to increase, it will continue to hold it for many generations to come. But, as

we in America can create more home calls for money, the manufacture of securities being one of our chief industries, than we can supply, there is no disposition among us to go abroad for investments, and we need not concern ourselves about the offerings made elsewhere.

State Obligations.

The unquestioned credit of "governments" has led many persons to the inference that the next safest and best investment securities are the obligations of the States, as they rank second only to the national authority. How far that impression is verified by the facts of the case will appear from the statement that, of about twenty of the thirty-eight States of the Union that have out

sufficient paper to test their dispositions in the matter of paying debts, only eight or nine faithfully keep all their agreements, and as many as twelve, being nearly one third of the whole number, are in default as to all or portions of their outstanding obligations; that the State bonds that are promptly provided for amount to a little over $100,000,000, while the principal of those that are defaulted on or have been in default (some of them compromised by the substitution of issues of reduced amounts) is not far from $200,000,000; and that the sum total of the delinquency, when unpaid interest is added, exceeds $300,000,000.

It is true that a considerable portion of the last-mentioned figure is made up

of what are known as "carpet-bag bonds," having been put out by States that had been in rebellion during the process of their reconstruction, and which are repudiated on the alleged ground that the governments creating them did not fairly represent the people of the States. But that is by no means true of all. Virginia owes the largest amount, and the whole of her indebtedness was created before the war. The same thing can be said of Mississippi's debt and of the most of that of Tennessee and some of the other delinquent States.

When the reasons for their failure to maintain their credit are sought for, the most potential and obvious is found in the fact that States, under the na-

tional Constitution, can not be sued or otherwise legally proceeded against on account of their pecuniary undertakings, at least to private parties, and that consequently there is no way to compel them to pay if they are indisposed to pay. Some of them have shown that they are not above taking advantage of the situation. This is the more to their discredit, because it was not the intention of the Constitution's framers that they should enjoy any such exemption. The Constitution originally gave a right of action against the States. The eleventh amendment, which inhibits the suing of States by private parties, thus taking away a right previously existing, was meant to cut off certain claims growing out of the Revolutionary War

and for no other purpose, no one at the time of its adoption dreaming that it would be used by the States to shield themselves from subsequently contracted liabilities. The States that thus avail themselves of it are really guilty of a double fraud. But the fact that so many of them have not scrupled to do so, shows the danger of trusting to their pecuniary promises.

Whoever buys the paper of a State should do so with the distinct understanding that he has nothing but its honor to rely upon, unless the commercial relations of its citizens should be of such a character as to make its financial credit important to their business interests. There is for that reason little likelihood of such States as New York

and Massachusetts ever repudiating their obligations. But when such conditions do not exist, the faith of a State is an uncertain dependence. The public conscience in the matter of commonwealth obligations is notoriously lax, what is everybody's duty being looked upon as no particular individual's; and such a thing as State pride, except in the form of prejudices and antipathies, which are very likely to be directed against public creditors, can scarcely be said to exist. Of all the States, Virginia has professed the highest regard for her reputation, yet she has been the most brazen of repudiators.

How easy it is for a State to ignore its contracts without attracting public attention to its conduct is strikingly

shown in the case of Texas, which is not one of the twelve bond repudiators referred to. Texas has issued, and now has outstanding, agreements to furnish land from her public domain calling for about eight million acres, an area nearly three times the size of Connecticut, which she does not supply to the certificate holders. No charge has ever been made that the warrants were illegally issued by the State or improperly obtained by those owning them. The State had the land when they were created, and could without difficulty have met her agreements; but, after the warrants were out, she proceeded to give away such extensive areas for the building of a State house, for the endowment of universities, for the pensioning of

wounded Confederate soldiers, and for other local purposes, that the holders of her land scrip can get no satisfaction. As matters stand, the State has the consideration for the land, but those who have her paper can get neither land nor money; and although this condition of things has existed for several years, during which there have been several sessions of her legislature, Texas has taken no steps to redress the wrong she has done, and there seems to be very little desire on the part of either officials or citizens that reparation should be made or concern about the disgrace that attaches to them and to their State.

A very little reflection suffices to teach the unreliability of obligations

resting upon no more substantial support than public opinion. Divided responsibilities are always lightly esteemed, while no burdens are so sharply felt as those imposed by the commonwealth. Tax-payers will shrink from public debts, who would never think of shirking individual liabilities. National pride and the other elements of which patriotism is compounded, even in cases where there may be no pressure from the governments of foreign bond-holders, may, and probably will be sufficient to protect national agreements, but the undertakings of subordinate communities need more positive backing. It does not follow that there will be no repudiation because of an honest purpose in their contracting. There is always

the possibility of political or other disturbances that may lead to revulsion of sentiment or give demagogues the opportunity to assail the public credit. We are not without illustrations in this country. Virginia's bonds at one time commanded higher prices than those of the Federal Government; yet they are to-day selling at figures far below par, and some of them, or the certificates representing them, can be purchased for a few cents on the dollar. That they and State debentures generally, when all the contingencies attending them are considered, should bring as liberal prices as they do, is the only thing about it that is surprising. Such securities, in the writer's judgment, are, at present quotations, beyond all question, the

dearest investments we have in the market.

Municipal Obligations.

What has just been said concerning State undertakings is, in part, true of these. How ready our minor communities are to create debts, and then ignore them, has been proved by very many illustrations, some of them of recent date. With the revival of general business at the close of the war, came a period of rapid development in some of the newer parts of the country. Numerous railways were projected, and municipalities were called upon to aid in their construction by the donation of bonds. The market was soon flooded with them, particularly from the States

of Illinois, Missouri, Kansas, Arkansas, and their immediate neighbors. But the enthusiasm that led to their creation was quickly succeeded by disgust when the burdens they imposed began to be felt. Many were repudiated, and, but for fear of the courts, that would have been the case practically as to all. In Missouri a Repudiators' State convention was held, and resolutions denying all moral responsibility on account of municipal issues, and advising general legal resistance, were published to the world. Nearly one half of the indebted cities, counties, and towns of Illinois and Kansas denied their liability, and in Arkansas repudiation was universal. Nor were there lacking such examples in the East. Very many investors who had

put their money into the rejected bonds, became frightened, and sold out at whatever figures they could get. The aggregate loss was very great.

Nor are such cases at an end. The writer has before him, as these pages are being prepared, a number of certificates of indebtedness recently put out by one of the best-known and most wealthy counties of Colorado. One of the warrants was issued in payment for furniture for the local court of justice, several for salaries of county officials, a number for witness-fees, etc. All were created for purposes entirely proper and necessary, and yet payment is absolutely refused.

As a general thing, repudiated municipal undertakings, being sustained by

the courts, have in the end been paid or compromised; but many of them have been defeated on technical grounds, and proved wholly worthless in innocent purchasers' hands. The wealthy capital of the State of Kansas, to the State's disgrace as well as her own, has to-day outstanding $100,000 of bonds which she does not pay because the courts have held that she can not be legally compelled to pay them! Everybody has heard of the case of Memphis, Tennessee, which, in order to escape proceedings for the enforcement of her obligations, got her charter repealed, and ceased to be an organized city.

What has been said concerning the moral strength, or rather moral weakness, of State bonds, will apply with

equal force to those put out by municipalities. The people in the one case are no more honest than in the other; and local pride, except on the part of a few leading commercial cities, in such matters counts for very little. Nevertheless, as will be seen, municipal obligations have an important element of security that State issues do not possess. Their collection, provided they are properly created, can be legally enforced. But this fact should be accepted for no more than it is worth. While the bulk of the paper put out by our municipalities is undoubtedly good, the only safe course in buying it, in very many instances—possibly a majority—is to proceed upon the assumption that payment will not be made if it can be avoided,

and to graduate prices as well as adopt precautions in accordance with that view. The aspect of the matter presented is not particularly charitable or flattering; but, whoever invests his money on a different one, runs an excellent chance of losing it.

Hence, in taking municipal securities, it will be seen how important it is that all points affecting their validity should be inquired into. One of the first questions to be settled is whether they have been the subject of litigation, and what, in that case, has been the result. Another is whether they belong to the original issue, or have been put out in settlement of an earlier and disputed series. Compromise municipal bonds have generally been good, al-

though there have been instances in which even they have met with opposition from their makers. Fortunately for later buyers, the litigation affecting municipal undertakings has been so extensive that nearly all questions that can possibly involve their validity have been passed upon by the courts, and it is not, therefore, now a difficult matter to determine, if reasonable prudence is exercised when such securities are offered, whether they will stand the legal test. But, of course, legal considerations are not the only ones to be studied. The pecuniary responsibility and general credit of the contracting communities should not be overlooked.

The importance of the position among investments that is held by mu-

nicipal issues is apparent as soon as we consider the formidable sum which, in the aggregate, they make up. In 1880, according to the census taken that year, they amounted to $871,507,373. But, enormous as that figure appears to be, it was considerably short of the actual total, because very many issues were then repudiated or in litigation, and, their validity being denied, they were not reported to the census-takers by the makers of them as any part of their debts. The most of them have been sustained, and with them and the additions that have since been made in new undertakings, the present aggregate would doubtless considerably exceed a thousand million dollars. A great deal of investment capital is, therefore, in

them. In the main it is well and profitably placed. As our cities and other municipalities grow in wealth and population, they become better able to take care of the paper they have out; particularly as their debts of late years have not increased as they formerly did. A number of the States have by constitutional enactment prohibited their communities from making donations of bonds to railroads and other corporate enterprises, and others have so restricted them that the worst abuses of the practice are no longer possible. On the whole, if proper precautions are taken by the buyers, municipal paper may now be looked upon as one of our safest and most remunerative investments.

Railroad Mortgages.

We have now come to a class of securities in which by far the greatest amount of money has been invested; much of it permanently, as the class under consideration has been the graveyard of a vast aggregate of capital. An idea of the volume of interest-bearing obligations predicated upon railway property in this country can be formed when it is known that, at the time these pages are written (1887), our operated railroad lines, all told, foot up not less than one hundred and thirty-three thousand miles. The portion that is unencumbered is so small as to be scarcely worth considering. The average funded indebtedness is about $30,000 per mile,

or, in all, $4,000,000,000. But that figure, enormous as it is, by no means tells the whole story. So many railroads have been foreclosed and sold under one series of bonds, that they might be used in reorganization proceedings as a basis for another issue—the process in some cases being several times repeated—that it would probably be no exaggeration to say that the liability, one time or another created, would average nearly $50,000 per mile, or a grand total of $6,000,000,000.

As our railways, with few exceptions, are valuable and productive properties, they furnish an excellent security for interest-bearing paper, provided always that they are properly managed and not overburdened with debt. There

are at all times good railroad bonds in the market; the trouble is, that there are poor ones there also. How is the investor to discriminate between them? When the mortgages are on roads that have been in operation for considerable periods, there is little difficulty in reaching safe conclusions. The principal question involved is that of earning capacity, considered in connection with amounts of funded liability, and our railroad manuals furnish all needed data on those points. Of course, there is always a danger from rival enterprises in the shape of "paralleling" lines, and possibly a greater one from speculating directors, who are willing to wreck the properties intrusted to their charge; but the peril is small in comparison

with the interests involved. The puzzle is when it comes to the paper of new roads—of roads that are in course of construction or just completed—and generally by far the greatest amount of railroad bonds offering in the market is of this description. Here the earnings guide-board is lacking. By what rule or rules is the investor in such cases to govern himself?—because many of the new securities are good, and as they can be had at lower figures than old ones, when properly sifted from the others they offer bargains that should not be neglected. The answer involves a brief description of the prevailing methods of railway construction.

It is quite within bounds to say that the principal incentive to the building

of railroads nowadays is the expectation of making profit out of their construction. The time for making roads at a sacrifice to their projectors, and because they are wanted by the public, has gone by. Railroad-building is now a regular industry—as much so as the erection of houses or the manufacture of machinery. When a new railway enterprise is undertaken, its authors expect to make the road not only supply funds for its own construction, but to leave them a handsome balance over. This is accomplished by mortgaging the property to that extent that the bonds can be sold for more than it will cost. In that case they are very likely to exceed in amount, although not always necessarily so, the security on which

they rest; in other words, be composed, in good part, of "water." Default in the payment of interest, and the bankruptcy of the enterprise, are the usual consequences. What follows is matter of routine. The property is put into the hands of a receiver, the securities are effectually discredited—perhaps bought up by a syndicate of crafty speculators—and then the work of reorganization is entered upon. The reorganizing of broken-down railroads has become another regular and profitable business. Calculation is made to determine on how many obligations the property can earn interest, and it is arranged to reissue for that amount, the old securities being proportionately exchanged and retired. Their holders, who are usually

glad to save something out of the wreck, are expected to accept bonds for a smaller face value, and probably at a reduced rate of interest, and at the same time meet a cash assessment to cover the cost of the operation. They not only have the water squeezed out of their holdings, but they have to pay for the squeezing.

The lesson to be drawn from the foregoing facts is easily stated. Let the investor, before he purchases, ascertain whether the bonds offered him are secured on a road that has gone through the reorganization process; and if not, unless he feels that he is compensated for the risk he takes by a great reduction in price, he had better keep his money, or find some other place for it.

Reorganization is almost certain to come sooner or later. It is the rock ahead. There are exceptional cases, of course, and if the investor has the means of making a thorough investigation, he can take advantage of them; but he ought to be very sure of the ground before venturing upon it.

Railroad Stocks.

Of course, many of these are good, and a great deal of money is profitably invested in them; but, as a rule, railway bonds, while at times possibly not yielding as liberal returns, are much to be preferred by ordinary capitalists. There are many things endangering stocks that do not threaten mortgage bonds. The latter are secured upon

property that is permanent, but the value of stocks rests upon management that is subject to constant change. The ambition of railroad directors is even more to be dreaded than their dishonesty. How many magnificent railway properties have been ruined by the suicidal policy of extending roads, adding branches or taking on leased lines! More particularly has this been the case when their stocks are listed at some exchange, and their managers happen to be operators there. The temptation to increase or diminish quotation values, for speculating purposes, is then a constant menace to conservative shareholders. The fact that a stock pays dividends, which is the only thing that the most of investors look to, is no cer-

tain criterion of value unless the control of the property can in some way be satisfactorily guaranteed.

Other Stocks.

Of these there is no end. The tendency of business, owing partly to an abundance of capital and partly to increasing competition, is to a large scale of operations, if not to absolute monopoly. One consequence is, that it is more and more passing from private parties into the hands of corporations. Corporations issue stocks, and sometimes that is one of the leading purposes of their formation. Some of the stocks are good and a good thing to have, and others are wholly worthless. No rule, beyond the exercise of common sense,

and the use of diligence in acquiring reliable information, can be given for their selection. Some dangers, however, can be pointed out. The greatest of these is the temptation of large dividends. Investors are altogether too prone to accept present realizations as evidences of future profits. It may be taken as a rule that, in this age of plentiful money, no legitimate business that is open to public competition will long pay exceptionally well. The greater its earnings at first, the stronger will be the competition in the end. Hence we find that many manufacturing companies, especially in New England, that made handsome returns to their shareholders for considerable periods, have gradually ceased to pay dividends at

all, or been forced into liquidation. If not strictly a legitimate business, anything in the nature of profits must be deceptive.

When corporations enjoy monopolies by virtue of patent rights, their earnings have often been very great, and early investors have made splendid fortunes. Those who come later, however, take chances that are too frequently underestimated. There is always the danger of new discovery. Gas is threatened by electricity, the telegraph by the telephone, and even steam is menaced by new-fangled motors. Turnpikes and canals have been superseded by railways, and old railroads may at any time suffer from the competition of new ones. Another fact which is too often lost sight of

is that investing in a patent right, or in a business founded upon it, is like investing in a mine that is being constantly worked out. Every hour shortens its monopoly. But the greatest danger in such cases is over-capitalization. Such stocks nearly always contain too much water, and water invariably tends to lower levels.

In handling shares the highest art is in selling rather than in buying. That is something that the most of investors do not understand. They hold on too long. When they have a good thing, they infer that it will always remain so, and accordingly retain it until its value has departed or greatly deteriorated. Stocks require constant watching. If any one wants to go to sleep on his in-

vestments he had better select mortgages.

While it is true that numerous stocks are very much like lottery-tickets — a good many of them, and among them the shares of mining companies, from the very nature of the case—and dealing in them is a species of gambling, the observation is not true as to all. The law watches over some of them in a way to protect them from the most obvious dangers. For that reason, bank-stocks can generally be relied upon as representing actual cash investments. In some of the States the same thing is true of insurance companies' shares. Efforts have even been made, in certain quarters, to apply the rule to all corporations; but, as the corporation laws

of other States are generally open for anybody to organize under, such restrictions have little practical value. In acquiring stocks, it is always well to inquire—a point generally overlooked—under what statutes the companies have been organized. If they are found to exist by virtue of the corporation laws of States that are notoriously lax in their requirements—especially if their corporators are residents of other States—the presumption is against them. There is pretty certain to be bad faith in their inception, and a bad beginning is likely to make a bad ending.

Farm Mortgages.

We have now reached a class of securities that has been steadily grow-

ing in popularity, and more and more has absorbed the capital of investors. That it must have merit is a fair inference. All things considered, it is doubtless the best now offered on a large scale. Land in this country is steadily appreciating, except in a few of the older sections. There is no reason to apprehend, at least for years to come, a turn in the tide. Hence, paper resting upon landed security, especially in the newer and more rapidly-developing districts, if properly graduated in amount, is almost certain to be good. What is required is sufficient care and judgment in placing the loans; and so complete is the system applied by some of the parties that make a business of putting mortgages upon land, in apportioning their

money allotments to the market values of property, that they can be accepted with even greater confidence than if arranged by capitalists themselves. Hence many savings-banks and other institutions of a fiduciary character, as well as numerous conservative individual investors, have put their money into paper secured on farming-lands, and lands which they have never seen, and had no opportunity to see.

At the same time it must be remembered that, if the strictest good faith is not observed, there is here a dangerous opening for fraudulent imposition. Should there be collusion between the loaning agent and the land-owner, the advance may be in excess of what is justified by the value of the realty, or

the title to the latter may be defective. The surprising thing is, when the popularity of farm-mortgages and the extent to which they have been dealt in are considered, that cases of deception have not been more numerous. As it is, they have not been wholly wanting. The writer once had occasion to visit what was called a farm, and as such had been made the basis for a loan, in one of the newer States. He found a piece of prairie-ground on which there were signs of occupancy at some considerably anterior period; but which was then deserted, and to all intents and purposes wild and waste. It was worth no more than land in the same neighborhood, on which there had been no attempt at improvement, and that could be purchased for

less to the acre than the mortgage upon this property. The owner, having secured such a liberal accommodation through the carelessness or knavery of some loaning agent or dealer in mortgages, had pocketed the money instead of expending it on his place, and left the ground for his creditor to make the most out of that he could. Competition among professional money-lenders in some of the Western States has become so sharp that such cases are not only possible but probable. Villages away out on the frontier, with scarcely sufficient population to make respectable settlements, are often the sites of investment companies—sometimes of several of them—doing business under most pretentious names, and offering amounts

of paper that, in view of the surroundings, are truly marvelous. Nothing which, by the most liberal interpretation, can be called a farm, seems to escape them. The writer was not surprised, when told by "one of the natives" of a border community, that there was not a hen-coop in his part of the country that did not have a mortgage on it!

One of the devices of these enterprising companies is to offer their own guarantees as to both principal and interest of all mortgages negotiated by them. It is hardly necessary to ask what, in a majority of cases, such indorsements are worth. The writer knows of one company that claims a capital of only thirty thousand dollars,

and it is doubtful if half that amount were ever in its treasury, which, by its own profession, has disposed of indorsed mortgages amounting to several million dollars. The only safe policy, in buying paper of this description, is to deal with parties of known responsibility in the business, and that have reputations which they will naturally be most anxious to protect. There are a number of such concerns, and it is here unnecessary to mention them. Some of them are individual operators, some business firms, and some incorporated institutions with large resources that make a specialty in this line. It would not seem to make much difference which is dealt with, although when the loans have a good while to run, and the in-

terests to be watched over are at a distance, prudence would suggest, in view of the uncertainty of human life and the many vicissitudes that fortune has in store, that it would be wise to deal with parties having the elements of greatest permanency, and which are likely to be corporations.

It is scarcely necessary to add that, in selecting investments of this sort, there is a choice apart from the market values upon which they are secured. What is wanted is, not merely safety as to the principal of the loan, but regularity in the payment of interest. In sections where only one crop is depended on by agriculturists, such as wheat or cotton, there is greater danger of defaults from natural causes than in dis-

tricts where there is a variety of productions. Then, there is a difference in populations. Some have sounder notions as to financial obligations than others. It is hardly to be expected that individuals in communities that repudiate or neglect public debts, will be shining examples of personal integrity. It will always be well to give the go-by to repudiating districts. Apart from the general lack of sound business ethics in such localities, there is a reason why it is unsafe to trust the monetary contracts of their people, and that is the danger to be apprehended from unfriendly collection laws. When communities take to fighting creditors, they are not likely to confine their opposition to any one class. By means of exemption laws,

stay laws, limitation laws, assignment laws, etc., they can make it almost as difficult to collect from individual debtors as from the public, and, as a matter of fact, we find legislation of that sort most pronounced wherever commonwealth and municipal undertakings are least respected. For instance, Texas, in selling her lands to outsiders, and then converting them to the uses of her own institutions and citizens, as heretofore described, has shown an utter disregard of all moral obligations, and here it is that the laws have been apparently made with special reference to protecting local debtors from the clutch of outside creditors. In the most of the States a sealed instrument for the payment of money is good for twenty years,

but in Texas, if not sued on in four years, it is barred by "the statute"; and thousands of people having claims against Texans, have permitted them to be lost through ignorance of that peculiar provision. Again, in Texas a man may possess a great deal of valuable property and yet be execution-proof. In addition to "all provisions and forage on hand for home consumption," all household and kitchen furniture, all implements of husbandry, all tools and apparatus used in a trade or profession actually followed, family portraits and pictures, a gun, two horses and a wagon, a carriage or buggy, five milch-cows and calves, two yoke of work-oxen, twenty hogs, twenty head of sheep, all bridles, saddles, harness, etc., the debtor head of

of a family may hold a homestead, not in a town or city, of not less than two hundred acres of improved land, or in a city, town, or village, of lots of the value of five thousand dollars, "without reference to the value of improvements thereon," and which, of course, may be worth much more than the land. In Texas, to get moieties under assignments made for their benefit, creditors must surrender in full their demands—from which provisions it may not unreasonably be inferred that Texas is a much more healthy region for debtors than for creditors. In Colorado, whose repudiations have also been referred to, a promissory note may be "outlawed" by two years' indulgence on the creditor's part, and a judgment of a court of record is

good for three years only. The moral from such cases is obvious enough. Keep your money out of Texas, Colorado, and other repudiating sections if you ever want to see it again. The higher rates of interest to be obtained in such localities, and which are among the penalties for their treatment of those with whom they have dealing, will not compensate for the increased risk. Nor would it be imprudent to avoid cities and towns of tainted financial records, however prosperous they may appear to be, of which the capital of the State of Kansas, whose course is elsewhere considered, is, by reason of its wealth and prominence, a notable example.

Ranch Securities.

Akin to the obligations just treated of, because having a basis in real estate, and yet in so many points differing from them as to seem to call for their consideration under a different heading, are securities predicated upon ranch-properties. Cattle-raising, which, in earlier times, was looked upon as a part of the farmer's regular occupation, has of late become almost a separate industry, and grown to enormous proportions. Ranching, however, is not confined to cattle-production, as the term "cattle" is ordinarily understood. We now have sheep-ranches, horse-ranches, pig-ranches, goat-ranches, and even ostrich- and goose-ranches. The marvel-

ous growth of the ranch business, in quite recent times, has had the natural effect of bringing upon the market a class of securities that has absorbed, and unfortunately dissipated, a good deal of investment capital. For a time the business was remarkably profitable. Lands in the South and West, in large bodies, were obtained at nominal figures—often by mere appropriation—to operate them as ranches cost but a trifle; for, as a rule, neither houses nor fences were constructed or needed; and the prices of cattle, largely purchased for the purpose of stocking other ranches, were high and constantly advancing. Everybody in the business made money, or seemed to be making it. Speculation upon an extended scale was an inevita-

ble consequence. Stock-ranching companies—"stock" in the corporate sense — were soon being numerously organized with the usual concomitants of mortgage-bonds and share-certificates, and, of course, the public was invited to share in their benefits. Circulars that they put forth told very flattering tales. The lands which they possessed, often little better than desert wastes, were usually described as worth from five to twenty-five dollars per acre; their cattle were estimated at the highest market prices; and calculations were gone into to show, from the increase of their herds and the quantities of beef and young stock to be annually harvested, that they could not fail to be first-class bonanzas. The highest rates

of interest and most liberal dividends were promised. It is no wonder that a good many investors were captivated.

We all know what the result has been. Droughts in summer and frosts in winter, for which no allowance had been made, and against which no precautions were taken, wrought havoc with the herds, and production increased until the ranges were overstocked, and prices went down and down with growing competition until the end was failure and bankruptcy.

And yet it by no means follows that ranch securities will always be worthless. Stock-raising is an entirely legitimate calling, and will be successfully prosecuted on a large scale, upon our Western plains and prairies, which are

by nature intended for that particular use. After such arrangements have been made for shelter and feed in the winter, and for water in the summer, as ordinary prudence dictates, and the business is brought to that stage where it will no longer be conducted by "cattle-kings" and "cattle-queens" as a venture to be decided by the chances of the weather and the market, but by ordinary workers, content with moderate profits, and mindful of all the opportunities and economies, ranching will be a reasonably safe basis for money investments of all kinds.

WATER-WORKS LOANS.

These are becoming very plentiful—how plentiful may be inferred from the

fact that a directory of water-works for towns and cities in the United States, recently issued by a New York publisher, shows that of over fourteen hundred such establishments, the number of those owned and conducted by private (although incorporated) companies exceeds those belonging to the municipalities by over one hundred and fifty. The communities served by these companies range from villages of less than two thousand population up to a city of one hundred and twenty-five thousand souls; and the debts of the companies, in nearly all cases represented by coupon bonds, run from twenty-five thousand up to four million dollars. As the stock issued by the companies fully equals their bonds, it will be seen that

the aggregate of securities put out by them is very great.

The building of water-works has become a regular business, and has been followed with very satisfactory results by contracting parties that, in many instances, are owners of or interested in the factories producing the pipes, pumps, and other materials and machinery necessarily used. The customary method has been to secure, in the name of an incorporated company, that is merely a cover for a professional builder, a contract with a city or village by which it is to be supplied with water for public uses—protection against fire being the principal one—at a stipulated annual price, or for so much per hydrant, with the privilege of selling

water at specified rates to as many private takers as can be found. The conditions vary, but a supply of good, wholesome water, in adequate quantities, is always one of them. A failure on this point is to invalidate the contract, and there may be other grounds of forfeiture.

For the money with which the works are to be constructed, the builder ordinarily depends, sooner or later, upon the sale of bonds. He even expects to have a balance left from their proceeds when the work is done, and that, with the stock of the company, which may or may not have a value, is to be his profit. The works, in other words, are relied upon to build themselves, and pay the enterprising projector besides,

there being in this respect no difference as to the construction of railroads and water systems.

The writer has before him a pamphlet put out by a dealer in investments, in which water-works securities are especially commended. "In no other class of mortgages," says the pamphlet, "can so many and such strong elements of safety be found, since in the whole history of water-works there is but one known instance of the foreclosure of a first mortgage (where the works were completed and in operation), and in that instance not a dollar of loss was sustained by the bondholders." While the author of the foregoing could not have been familiar with the case of Memphis, Tennessee, in which the unfortunate

water-works bondholders received less than twenty per cent of their demands from the proceeds of the foreclosure sale, nor of those of Galesburg, Illinois, and other bankrupt companies, there are, nevertheless, reasons why water-works mortgage-bonds, if issued under proper conditions, should be among the very best. They are secured on property (underground pipes, etc.) which is but little exposed to fire and accidental injuries, and the business upholding them is usually without competition, conducted wholly for cash, and in growing communities must of necessity increase. But there are perils and drawbacks. There is here, as elsewhere, the possibility of overbonding. A greater danger is cheap construction. In no

class of works, perhaps, is there an equal liability in this direction. The most of the material used is in pipes that are buried and out of sight. It is possible to employ an inferior article, such as sheet-iron instead of cast-iron, which will temporarily answer the purpose; and as the builder's interest in the works, after their acceptance by the community with which he has contracted—usually not as critical as it should be—and the sale of his bonds and stock, is at an end, the temptation besetting him is very strong. Five or six years is the limit to the probable use assigned by engineers to pipes and other materials in works that the writer knows of, that are covered by mortgages authorizing bonds having from twenty to twenty-five years

to run. If, at any time before the bonds mature and are satisfied, it is found necessary to use the funds needed for principal or interest, in the reconstruction of the property, it is easy to see how the bondholders may suffer. Hence to them the very great importance of knowing, before they part with their money, sundry points about the security they are getting, that the most of bond-purchasers rarely think of inquiring into.

Nor have we reached an end of the risks attending this class of securities. The question of water-supply is always of first importance. Water-works without water, and plenty of it, are, of course, of very little account. Now, it is very easy to understand how difficulties on this score may arise and increase.

When works are first constructed, there may be a sufficiency of water of satisfactory quality within easy reach, and the builder, anxious to get through with his work as quickly and cheaply as possible, that he may realize on it, will naturally avail himself of such supply and look no further. But it does not follow that it will always or for a considerable time be adequate. Running streams are usually depended on, and everybody knows how liable they are to become corrupted in a settled country, and, while the demand upon them is increasing their volume is likely to diminish. The writer knows of more than one company that on this point has met with embarrassments not dreamed of when its operations were begun. In one instance

the cost of a new water-supply, made necessary by the deterioration of a running stream, exceeded the entire original outlay. The point is one upon which, in every instance, not only are contracts made especially stringent, but a strict compliance is certain to be demanded. How many of our water companies will be able to meet their obligations in this regard, during the periods that their bonds are issued for, without the expenditure of considerable sums of money not at first anticipated? To provide a fresh water-supply may prove a more serious undertaking even than the substitution of a new plant for an old one.

When securities founded upon water-works in localities where they are actu-

ally needed, and erected under seemingly favorable conditions, are attended with such liabilities, it will be seen how much greater the danger must be when the works are located, by speculative builders, in communities that do not need them, and of uncertain ability for their support, purely as a basis for the manufacture of bonds and stock. The country has been ransacked for water-works sites, and plenty of works have been built in villages that the majority of our people have never heard of. They may grow and become important places, and they may not. The obligations created in such cases may be valuable, and they may not. But it is quite safe to say that, if the movement in this direction continues and goes much fur-

ther, the day is not far off when we will be favored with a plentiful crop of defaults and water-works foreclosures.

STREET-RAILWAY BONDS.

Much that has been said concerning water-works securities is applicable to these. There is not the same opportunity to delude by the use of inferior materials, since the works are on the surface; but there has been an equal disposition to push construction into territory of doubtful earning capacity. Many roads have been built where they were not needed, and where they can not pay—at least, for considerable periods to come; and more of that sort are likely to be built. At the same time, when advantageous locations have been se-

cured, with franchises properly covering the ground, and particularly if exclusive, few better properties exist, and the securities based upon them can be safely recommended. All the investor needs to do is to investigate thoroughly before he buys. He should satisfy himself that the road is in a community that is able to sustain it; if in a city of known standing, that it occupies streets likely to give it permanent business, and has not been built in remote suburbs or along country highways as a pretext for bond-making; that its franchise is sufficient to protect it from dangerous competition; that the securities have been legally issued; and, above all, that there are not too many of them.

Miscellaneous Bonds.

Much that has been said concerning water-works and street-railway mortgages is applicable to various other securities, notably those of gas and electric-light companies. In cities and villages, when the latter are of considerable size, streets must be lighted, and contracts for the purpose are always sought by those contemplating the construction of illuminating-works. Such contracts are usually for short periods, one year being the ordinary term, and for that reason there is greater danger from competition than in the matter of water-supply. Indeed, gas and electricity are likely to be competitors in most fields for a considerable time to come, and

the profits from their operations must be correspondingly abridged. Nevertheless, many of the securities predicated upon them should be good, and, if selected with care, can be taken with confidence. Of course, there may be defective construction in material and workmanship, and the buyers of gas-light and electric-light bonds would do well, before parting with their money, to inform themselves thoroughly on these points as well as others. In the larger cities there is less difficulty in estimating securities of the kind, as they are pretty certain to have local quotations that determine their values for the time being.

When it comes to bonds secured on manufacturing and other private proper-

ties, which may or may not be covered by articles of incorporation, no general rule can be laid down. Each interest must be measured by its own circumstances, and investors can best determine for themselves how far it is safe to trust the promises it makes. The markets in such cases are always more or less local, and those who see fit to put their money into them may be supposed to be in possession of all the facts necessary for the formation of correct judgments.

Mining Securities.

The best thing that, as a rule, can be done with these is to let them severely alone. There are good mining shares, but the average investor, who buys in-

to such properties, does not get that kind. It is a peculiarity of the mining business that it pays only when managed with exceptional skill, and that requires the property to be in the hands of its owners. Of course, the average investor can not take personal charge of the mines into which he buys, and would not know how to do it properly if he could. He is compelled to intrust his interest to the care of hired agents, and, if anything is settled in connection with mining, it is that about the only thing the ordinary salaried mining superintendent is good for is to draw checks on his employers, and bury their money beyond the reach of resurrection.

Bridge-Bonds.

When our larger streams are reached by railways, and must be crossed by bridges that cost a great deal of money, such structures have generally been erected by corporations independent of or separate from the railroad companies. As a general thing, being proximate to large cities, they have also been made to furnish the accommodations of public highways. Bonds and stocks issued upon them, while subject to many of the same vicissitudes, are quite as good as those of the average of railroad companies. No general rule can be laid down concerning them. Each security must stand upon its own merits, as the structure upon which it is issued must

stand upon its own foundations. There should in each case be an investigation before buying. One general caution, however, may not be inappropriate, and that is, to fight shy of the paper of companies owning very expensive bridges, unless the ordeal of a reorganization has been gone through. It is well known that the New York and Brooklyn Bridge, lying between the first and third cities in the country, does not begin to pay interest on its cost. The great St. Louis Bridge failed for the same reason, and it is doubtful whether such enormous structures can be erected at outlays that will justify them merely as business ventures and investments.

Substitution Securities.

Reference has already been made to an ingenious arrangement for issuing bonds upon the security of other bonds. The plan is comparatively a new one in this country, although something like it has long been practiced upon in England, and where more than one company adopting it has come to grief through injudicious investments. In itself it is simple enough. Certain persons having, as they suppose, superior facilities for investigating securities of one or more kinds, and sifting the good from the bad, organize with a view to purchasing such as they approve, and on the strength of which, when placed with some trust company or other safe

depository, they issue their own bonds for the market to the same amount, but usually bearing a lower rate of interest than those which they own. The difference in the interest and the lower figures at which they may buy, compared with those at which they sell, make their profits. To escape personal liability, and possibly to establish a higher credit, a corporation is formed, and the new obligations are produced in its name.

To such investors as lack confidence in their own judgments, or either do not have or prefer not to avail themselves of satisfactory opportunities for investigation on their own account, the system undoubtedly has recommendations. If securities were scarce and hard to find, it would be generally

popular; but, as long as they are plentiful, the majority of investors will prefer to trust to their own wits and save their money. And, after all, the judgment of one man may be no better than that of another, and those having money to dispose of, who, while relying upon themselves, exercise reasonable prudence, will doubtless, as a rule, enjoy the greatest peace of mind and occupy positions in many respects most satisfactory.

CHAPTER II.

*SPECULATING.**

NEW YORK has no more entertaining public exhibition than its Stock Exchange. It is one of the show-places of the city. The visitor who, for the first time, looks down from a gallery upon its members in the act of transacting business, is astonished at the apparent confusion he witnesses. He seems to have entered a mad-house. The idea that the market values of our leading securities should be determined by what appears

---

* The following chapter appeared in "The Forum" for October, 1886, under the title of "The Heart of Speculation."

to him to be a howling mob of incurable lunatics, is incomprehensible. But if nothing could be said against the Exchange, which is simply a big bazaar for the sale of bonds and stocks, except its tumultuousness and the seeming lack of dignity among its operators, criticism would have in it but an indifferent target for its shafts. Much graver questions grow out of its existence. Is it a harmless institution? Is it a public blessing? Is it a public curse?

As a great central mart for current securities, it would be unobjectionable. There is no reason why bonds and shares should not be publicly dealt in, and in large quantities, as well as dry-goods; as well as corn and cotton and beef and kitchen vegetables. If the

Stock Exchange was intended for or restricted to the *bona-fide* buying and selling of bonds and shares, not a word could be justly said against it. But is that its business? Unfortunately, no. Its chief occupation is wagering on stocks: its members, while going through the forms of buying and selling, simply bet their money, or somebody else's money, upon the rise or fall of the shares they select, as they would upon the shiftings of cards or dice. The Exchange, while having a share of legitimate business, is chiefly an immense gambling establishment.

Its members are divided into two classes—those who execute commissions for others, and those who deal on their own account. It is needless to say that

among the latter are the boldest and sharpest speculators of the day. The careers of these men can be sketched in very few words. Through the exercise of superior native wits or the accident of extraordinary luck, they flourish marvelously for a time; but only, as a rule, to lose their heads and their balance at last, and go down—often through a single disastrous transaction—faster than they went up. There are exceptions. Some flourish to the end, dying—generally young—or retiring with estates unbroken. But they are exceptions. Wall Street is a place where a few fortunes are made and a great many are lost. The stories of its magnificent triumphs, and of its equally magnificent wrecks, read like tales from the "Ara-

bian Nights"; some of them like passages from Dante's "Inferno." Wall Street has had its suicides by the dozen, and it will have plenty more. It would not be Wall Street without surprises. And yet there is a singular sameness in the ordinary broker's experience. He runs an exciting, if at times a rough and stormy, career, snatches or seems to snatch a good many pleasures by the way, makes and breaks with about equal *abandon*, wrecks his health in a hurry, dies early and suddenly, and then—well, then, when his affairs come to be settled, there are found to be large blocks of utterly worthless shares, perhaps a fast horse or two, a two-wheeled vehicle and trappings to match, some costly souvenirs, and very few solid assets, and the

business is closed in bankruptcy. Poor fellow, everybody has forgotten all about him!

Of the ordinary Wall Street speculator, however clever or however favored for a time, it is perfectly safe to say that, if he lives long enough and sticks to the business, he will finally come to grief.

But how about Vanderbilt *père*, who was more or less of a Wall Street operator all his many days, and a few other not wholly dissimilar if less conspicuous examples?

Ah! that brings us to a view of some of the interior workings of the New York Stock Exchange that the public has little conception of, and which alone will give a correct under-

standing of its real character. The popular idea is that the Exchange has upon its list, to be dealt in, all, or nearly all, prominent stocks and bonds of acknowledged value, impartially selected and solely because of their merits. There could be no greater misconception. We look there in vain for the shares of the Pennsylvania Central, whose stock has not a drop of water in it; of the Baltimore and Ohio, whose paper, notwithstanding some mistakes of its managers, is equally solvent; of the Boston and Providence, the Boston and Albany, the New Haven and Hartford, the Maine Central, and of dozens of other corporations whose management is unexceptionable, and whose securities are among the choicest investments. But if there

is a company with a speculating board of directors, and whose stock has been watered until it will float a respectable navy, its shares are pretty sure to be found on the Exchange's list. Or if there is a company that is absolutely controlled and directed by some particularly active and conspicuous manipulator, its stock may be looked for at the same place. There has never, apparently, been any difficulty in a big stock operator getting his issues upon the list. What has been the result? Simply that the most abominable rubbish has been unloaded upon the public.

Much, but not too much, has been said in condemnation of stock-watering; of the production of corporate certifi-

cates representing little or no cash investment, and which innocent persons are led to purchase in the belief that they are getting full values. But how is it that these fraudulent issues can be marketed, and the producers escape legal responsibility for the impositions practiced? Here is where the Exchange's work comes in. The Exchange is the conduit through which the water is safely carried into the investors' pockets. When it takes the stock upon its list, the Exchange becomes practically the seller, supplying the machinery and the means of transfer, and it guarantees nothing. Whoever buys at its board is understood to take all risks, no matter how much deception is used. He may be utterly victimized—often is so—but

he has no redress. Here is the medium through which the over-issues have been marketed. But for the Exchange's instrumentality, the facilities it has furnished, those stupendous stock-watering frauds which have become historical never could have been successfully consummated.

Once on the Exchange's list, there has never yet been a stock so worthless that, with a shrewd manipulator behind it, it could not be unloaded. The process has been a simple one: First, there are "washed"—singular how the idea of water runs through all stock operations—or prearranged sales of the stock. Outsiders are then told that there is money in it, and they begin to buy. The stock is duly "supported"—an indispen-

sable precaution—that is, it is taken at quotation prices when offered by outside owners, and so up and up it is marked, the speculative public taking large blocks in the belief that it is going higher, and with little thought of its actual value, until there comes a time when, the original supply being exhausted, the shares are no longer supported, and down, down they go. The real value of the stock has little to do with its negotiation. In the light of that explanation, there is no difficulty in comprehending how certain great railroad magnates, who are leading operators in Wall street, have amassed such colossal fortunes. They have been stock-manufacturers as well as stock-dealers. The New York Exchange has been their

field of operations—their market-place. Through it they have sold their wares. Had they, like ordinary speculators, confined themselves to other people's goods, it is questionable whether they would have grown exceptionably rich. They might have become poor, as the most of their associates have done. But when, with consciences conformable to their opportunities, they had the means of selling water at high figures and in practically unlimited quantities, it is no wonder that their fortunes swelled to fabulous proportions.

A glance at the Exchange's list tells the whole disgraceful story. What a column of tatterdemalions it parades! It looks as if, in making up its assortment, the listing committee had gone into the

highways and by-ways, with orders to bring in the lame, the halt, and the blind. Wabash is there, Denver and Rio Grande is there, Hocking Valley is there, Texas and Pacific is there, Bloomington and Western is there, Nickel Plate is there, West Shore is there, the whole noble army of frauds that once flourished so magnificently and bled the public so profusely, is there. Consolidated Gas, with thirty-five millions of stock that, from official investigation, would appear to have been evolved from a cash investment of less than twelve millions, is there, of course. It is a new accession, and shows how naturally inflated and adulterated securities seek the Exchange's forum, and how readily they are admitted. The Exchange has a com-

mittee to pass upon applications for listing, and which, in theory, excludes unworthy issues. It is supposed to act as a sieve; but certain it is that, sieve-like, it is no obstruction to the passage of water.

While its bond-list, as a whole, is much more respectable than its stock-list, it is noteworthy that the Exchange's dealings are principally in the speculative issues—the second and third mortgages, the incomes, the land-grants, and other junior or discredited securities. These are the driftwood of the market, which nobody buys to keep, because, yielding little or no income, they are of no account as investments; and they are bid up or bid down, according to the course of speculation at the time. When

actual investors wish to buy, as a rule they go to bankers and dealers who have nothing to do with the Exchange and pay very little attention to its quotations.

Oh, how gayly the business of making and marketing securities was but recently going forward in Wall Street! The inflation period that followed the depression from 1873 to 1879 was the golden era of stock speculation. The Exchange fairly rioted in profitable traffic. The public was supposed to be crying for shares, and the magnates of that institution were doing their best to meet the demand. They succeeded pretty well. Some of them built new roads and stocked and bonded them for only four or five times their actual cost.

They sought strange fields for their ventures—in the wilderness, upon the desert plains, among the mountain-peaks. They overleaped the national boundaries and rushed pell-mell into Mexico; and when all available openings were filled, they entered upon the work of "paralleling"—constructing new roads by the sides of old ones. The purpose of it all was the production of paper to be dealt in "at the board." It was bonds, bonds, bonds; stock, stock, stock; water, water, water. Millions upon millions of so-called securities were manufactured, costing little more than the blank paper upon which they were stamped and the mechanical labor bestowed upon it, and dumped into the hopper of the Exchange, to be by it

stirred up and turned over a few times, and then systematically worked off on the great investing public.

Very well does the writer, as well as a good many others, remember what it all came to; how on a bright day of May, of the year 1884, pandemonium, in the form of a panic, entered Wall Street; how the great throngs that gathered there and filled all available spaces surged and seethed like troubled waters; how great bankers and leading business men ran wild-eyed and bareheaded through the streets; how mobs of half-demented people crowded round brokerage-houses, richly-dressed women and gray-haired men among them, weeping and wringing their hands; how surrounding the doors of suspected banks

were groups of idlers who, with the true instinct of Wall Street denizens, were betting their money on the length of time their doors would remain open; how about the remorseless "tickers" in brokers' shops were gathered crowds of excited men, tremblingly watching the course of stocks that seemed to be going down, down to perdition, the water they contained suddenly turning to hydraulic pressure to crush them. Those who saw that spectacle in all its grim and terrible seriousness will witness nothing to match it this side of the "Inferno." Ah! the New York Stock Exchange then gave its patrons a treat many of them will not soon forget.

But the Exchange, with all its shortcomings, is at least useful if not neces-

sary in supplying quotable values and giving stability and tone to the business of the country. Is it? Let us see. On the 1st of March, 1884, Delaware, Lackawanna and Western stock sold on the board at 133⅛. In January following it brought at the same place only 82⅝. The next December it was up to 129⅝. "Lackawanna" is an old, conservative company, lightly capitalized, with an established business; a regular, uniform dividend-payer. Intrinsically the value of its shares has not varied in the past five years. "St. Paul" is another stock that, apart from speculation, should not change. Yet in 1883 it sold "on 'Change" as high as 108½, in 1884 down to 58¼, and in 1885 was back to within one per cent of par. In one half hour

during the May panic of 1884 the securities on the Exchange's list shrank in quotations over $100,000,000; in one day nearly $300,000,000. Not much stability, not much reliability there! Either prices had been much too high or they went much too low. The Exchange in the one case or the other, if not in both, failed to hold them at the proper level. Nor in this is there anything remarkable. Wide and sudden fluctuations are necessary results of the Exchange's methods. Its members are supposed to be divided between "bulls" and "bears"—those who try to advance prices and those who try to depress them; but all are as likely as not at one time to be bulls and at another bears. They have their stampedes. No drove

of cattle upon the Western prairie is more subject to sudden scares and erratic rushes than they are. Indeed, a wild herd of steers, with horns uplifted and tails in the air, charging across the plain, would give but a faint idea of the flurries and scurries of Wall Street bulls and bears in the midst of a round-up. When the market looks like going up, all hands are ready to lift it higher. When going down, all are ready to ride it to the bottom. The result is, a succession of extremes; and even when the entire Exchange is not blindly swayed to one side or the other, nothing is more common among its operators than the formation of pools to advance particular stocks or of combinations to raid others, artificial agencies in both cases being

freely used. How often, or rather how seldom, do Exchange quotations express the values that stocks would have if left to themselves or to the arbitrament of supply and demand!

In these things, as already said, to one familiar with Wall Street ways, there is nothing remarkable. It is upon fluctuations that stock speculation fattens. The delight of the regular Wall Street man is a wild market—the wilder the better. Quick changes bring him quick profits. He knows that a steady market means a dull market, and nothing does he more heartily detest. To him the most agreeable of all movements is that which sends up prices with a rush and a hurrah, creating what, in Wall Street par-

lance, is known as a "boom," and leading outsiders to purchase on the rise—of course, in the expectation of higher figures—and which then lets prices drop so suddenly as to shake or scare these purchasers out. In that way the broker gets both the money and the stocks, and the outsider gets a lesson. It is the theory of experienced operators, and undoubtedly a correct one, that the outside speculator rarely comes into the market until prices are up, and he can look back and see what he has lost by not venturing earlier; and is never so ready to sell as when prices are down, and he can look back and see what he has lost by not getting out sooner.

Instead of being a balance-wheel to the business of the country, the Stock

Exchange is far more likely to be a disturbing factor. It does not even furnish trustworthy news. Nowhere is it so difficult to get reliable intelligence concerning any stock dealt in there, as in Wall Street. The inventiveness of the speculative broker is something marvelous. He can ruin the country one hour and save it the next. He can blight the crops of a whole section, or he can fill the land with abundance. He can make war or he can make peace, exactly as his momentary interest demands. Rumor-mongering seems to be a part of his trade. He is the chief of liars. Perhaps he is the exception rather than the rule among his fellows —it is to be hoped that he is—but he is a pretty numerous exception, for all that!

What is the consequence? Simply that when a financial storm threatens the country, the Exchange is almost certain to be the center of disturbance. No other institution is so sensitive. It exaggerates all the symptoms of trouble. It sends out its alarming reports as the storm-cloud sends out its lightnings. Looking at it as the barometer of values, the timid naturally conclude that everything is lost, and thus the evil is unduly magnified. Wall Street is as much the natural field for panics as the prairie is for tornadoes.

If the Exchange has been of advantage to the business interests of the country, those who have had dealings with it should be ready to testify in its favor. Of the thousands and thousands

who have visited it in person or by proxy, and done a little business with it, how many are ready to rise up and call it blessed, except in a very qualified sense? If all were to give their experiences, what would the verdict be? It is to be apprehended that the evidence of a very decided majority would not be flattering to Wall Street's famous institution; that their testimony would be that they had found it easier to lose money there than to make it.

But why mince matters? Why deal in doubtful phrases? Why not at once declare what the discussion of the subject inevitably leads to—viz., that the New York Stock Exchange, which is the soul, the motive power of Wall Street, is an evil in the land, a danger

to private wealth, a disturbing force in general business, and a foe to public morals. A not overdrawn description would picture it as an enormous devil-fish with a hundred thousand arms reaching into all parts of the country, and all equipped with suckers more or less powerful, and busy every one of them, in extracting nourishment for the monster to which it belongs. The trouble is that its tentacles are rarely seen. They work in the dark; they have the gift of invisibility. But, oh, how many victims they have crushed! Yonder is a bank that is supposed to be as solid as the hills. Rich and poor make it the depository of their surpluses. It enjoys the confidence of all. But in an evil hour one of the arms of the Wall

Street octopus has fastened itself upon it and penetrated to its safe, and pretty soon its president, or its cashier, or its managing director, will be gone—gone to Canada — and the bank will be wrecked. There is a citizen who has the respect of all. He is a good man, useful in his community, and the stronghold of his family and his friends. But, somehow, he is caught in the deadly embrace, and soon he will be a bankrupt and a defaulter, if not a suicide. Such cases, by their frequency, have almost ceased to surprise; and yet they represent but a small portion of the losses actually wrought. Most of the injuries inflicted by stock-gambling are unknown, except to the sufferers. Wall Street's victims, as a rule, do not expose

their wounds unless they are mortal. The aggregate tax upon the country for the support of its operations is something enormous. It can not be otherwise when we see how Wall Street lives and flourishes. It maintains a good-sized army of operators, the membership of the Stock Exchange numbering nearly twelve hundred—without counting "curbstone" men and other camp-followers—who spend with the lavishness of soldiers of fortune, while some of them take unparalleled fortunes out of the street. And yet Wall Street does not produce a dollar. It creates nothing. It draws its sustenance entirely from outsiders. It is a blood-sucker.

That Wall Street should continue

to attract fresh patrons and victims, in view of the numerous warnings they have received, would be unaccountable were it not for that feverish desire for sudden riches which pervades the whole country, and which "the street" has been mainly instrumental in producing. It is said of the cuttle-fish that it discharges a fluid which darkens the water all about it, and so blinds its prey that they are helpless against its attacks. The Wall Street monster—the comparison still holds good—by the example of its few conspicuous successes and its general demoralization, so impregnates the atmosphere of the whole country with the speculative mania, that thousands and thousands can not resist it. There is no village so

small or so remote that it may not have its local speculator. No calling or profession escapes the contagion. The accommodations for all are ample. Wall Street has its wire connections with all points, and there are plenty of middlemen to instruct the uninitiated and take their orders for stocks. The "margin" feature is the cleverest bait. The fact that, by putting up one thousand dollars in cash, you can buy or sell from ten to twenty thousand dollars in stocks, and take a profit on the larger amount, is to many an irresistible temptation. Then, in theory, it is so easy to win by speculation! To buy at a low figure and sell at a higher, or to sell at a high figure and afterward buy at a lower, seems such a simple operation! It al-

most looks as if you could go into Wall Street and pick up money from the sidewalks. Those who have made the attempt, however, have found the practice very different from the theory. When the cleverest operators, the trained *habitués* of the street, so often make shipwreck, what hope is there for the inexperienced? A loss, however, is usually incurred before the real difficulties of the situation are realized, and then, in nine cases in ten, there exists on the part of speculators, out of sheer desperation or from the fascination that attends the game, the determination to try another chance, and in that way good money is thrown after bad until ruin is reached. It is folly to charge upon Wall Street sharpers the seduction of

such men. They lose because they want to make money, are not particular how they make it, and flatter themselves that they are sharp enough to win where others have failed. They are their own victims.

And yet they are not the only sufferers, and possibly not the greatest. The man who wins somebody else's money in Wall Street is far more than likely to lose it, and more with it, at the next venture he makes. And even the few so-called lucky ones who retire with their winnings, are not under all circumstances to be envied. The triumph of the man who victimizes the public with watered stocks, which are no better than adulterated wares or counterfeited coin, is not without alloy. He may re-

joice in the money, and in the fleeting importance it gives him, but he knows how he got his wealth, and he knows that others know how he got it. The sensitiveness of pride remains, even if conscience be dead.

But while the writer does not hesitate to arraign the New York Exchange, being the acknowledged center of stock speculation in the country, as an enemy to public morals and general business, he admits that it is not the only culprit of the kind. The Produce Exchange—or Board of Trade, as it is called—of Chicago, is a den of speculators, whose operations are even more pernicious. They affect more far-reaching interests. Stocks and bonds are in comparatively few hands, and these are generally strong

enough to withstand ordinary fluctuations. But the produce-gambler deals with men's necessities, he juggles with the staff of life. The soil-worker, who takes no part in the gamester's operations, and is in no wise responsible for them, is liable at any time to be robbed of his just rewards through their deals and pools; and the mechanic or other wage-earner, who is equally innocent of complicity with them, is compelled to pay them tribute on every loaf of bread and every cut of beef or pork he puts into his own or his children's mouths. Of all kinds of speculative gambling, that in breadstuffs and meats is the lowest, the meanest. The same comment, differing in degree only, will apply to such institutions as the Petroleum Board

of Pittsburg. Indeed, it runs the whole gamut of the speculative "exchanges" and "boards," from the highest down to the petit-larceny bucket-shop where, with a ten-dollar bill, you can purchase a chance on stocks, or oil, or wheat, or pork, or anything else that men gamble in. All are members of one family, and should be regarded and treated alike.

APPENDIX.

INVESTMENT SECURITIES.

Any one having a negotiable security is naturally anxious to know its market value; especially so if looking for a purchaser. If the security happens to be listed at a stock-exchange, it is an easy matter to get a quotation on it, and a buyer can generally be found at some price. Or, if looking for a not very common security, any one may possibly be aided in his search by reference to the same quarter. Few, however, outside of professional bond and share dealers, are familiar with the transactions of the exchanges, and it is probable that a preponderance of the securities dealt in at their boards are in the hands of people ignorant of the positions they occupy. To such parties, as well as to those who may be in quest of particular bonds or shares, without knowing exactly where to look

for them, and what their acquisition will cost, the following transcripts from the books of our principal exchanges, and showing the range of their operations, will be of interest, and sometimes of advantage.

It will be seen that, beyond giving, to some extent, a preference to obligations that happen to be strictly local or most generally held in the neighborhood, no positive rule of selection has been observed in making up the lists. The good and the bad are mingled in a way that is quite indiscriminate, and which to the casual observer must be somewhat bewildering. To any one, however, who has read the accompanying chapters there will be no particular mystery about it. Indeed, it is partly to illustrate the points therein made that the following record is given.

As prices bid or paid are constantly fluctuating, there is no use in giving present or recent figures; but any one interested in any of the securities on the lists can easily inform himself by applying to the proper quarter or quarters.

NEW YORK STOCK-EXCHANGE.

GOVERNMENT SECURITIES.

			Amount.
U. S. 4½,	registered	1891	244,251,000
4½,	coupon	1891	
4's,	registered	1907	737,812,800
4's,	coupon	1907	
6's,	currency	1895	3,002,000
6's,	"	1896	8,000,000
6's,	"	1897	9,712,000
6's,	"	1898	29,904,952
6's,	"	1899	14,004,560

STATE SECURITIES.

ALABAMA,	Class A, 3 to 5	1906,	6,728,800
	Class B, 5's	1906,	539,000
	Class C, 4's	1906,	959,000
	6's, 10–20	1900,	960,000
ARKANSAS	6's, Funded	1899–1900,	3,000,000
	7's, Little Rock & Fort Smith		1,000,000
	7's, Memphis & Little Rock		1,200,000
	7's, Lit. Rock, Pine Bl. & N. O.		1,200,000
	7's, Miss., Ouachita & Red Riv.		600,000
	7's, Arkansas Central		1,350,000
GEORGIA	7's, gold bonds	1890,	2,000,000
LOUISIANA	7's, consolidated	1914,	12,039,000
	7's, consolidated, stamped 4's		
	7's, consolidated, small bonds		
MICHIGAN	7's	1890,	231,000
MISSOURI	6's, due	1887,	3,242,000
	6's, due	1888,	3,251,000
	6's, due	1889 or 1890,	1,105,000

		Amount.
MISSOURI	Asylum or University, due..1892,	401,000
	Funding bonds, due....1894–'95,	1,000,000
	Hannibal & St. Joseph, due 1887,	1,000,000
NEW YORK	6's, gold registered.....1887,	942,000
	6's, coupon..... 1887,	643,200
	6's, loan.....1891,	4,302,600
	6's, loan.....1892,	2,000,000
	6's, loan.....1893,	473,000
NORTH CAROLINA	6's, old.....1886–'98,	4,738,000
	April and October.....	3,639,400
	to N. C. R. R., 1883–'4–'5, 7 coupons off..... April and October..... 7 coupons off.....	3,000,000
	Funding act...1866–1900,	2,417,000
	Funding act...1868–1898,	1,721,400
	New b'ds, J. & J..1892–'98,	2,383,000
	New b'ds, J. & J., A. & O.,	495,000
	Chatham Railroad.....	1,200,000
	Special tax, Class 1..... Special tax, Class 2..... Special tax, to W'n N. C. R. Special tax, Western R. R. Special tax, Wil. C. & Ru. R. Special tax, W'n & Tar. R. Trust certificates.....	11,360,000
	Consolidated 4's.....1910, Consolidated small.....	3,620,311
	6's.....1919,	2,593,000
RHODE ISLAND	6's, coupon.....1893–'94,	1,372,000
SOUTH CAROLINA	6's, Act March 23, 1869, non-fundable.....1888,	5,965,000
	Brown consol'n 6's...1893,	4,280,000

		Amount.
TENNESSEE	6's, old.....1890–'2–'8,	4,397,000
	6's, new bonds....1892–'8–1900,	
	6's, new bonds, new series..1914,	
	Compromise, 3–4–5–6's....1912,	2,014,000
	New settlement, 6's.....1913,	827,000
	Small bonds.....	51,600
	New settlement, 5's.....1913,	349,000
	Small bonds.....	10,300
	New settlement, 3's.....1913,	10,743,000
	Small bonds.....	350,000
VIRGINIA	6's, old.....	9,427,000
	6's, new bonds.....1866,	700,000
	6's, new bonds.....1867,	466,000
	6's, consol. bonds.....	20,239,000
	6's, consol. second series	2,442,784
	6's, deferred bonds.....	12,691,531
	6's, deferred bonds, trust receipts.	
DISTRICT OF COLUMBIA	3–65's.....1924,	14,033,600
	Small bonds.....	
	Registered.....	
	Funding 5's...1899,	943,400
	Funding 5's, small..	
	Funding 5's, reg'd..	

RAILROAD STOCKS.

Albany & Susquehanna.....	3,500,000
Atchison, Topeka & Santa Fé.....	68,000,000
Atlantic & Pacific	25,000,000
Beech Creek.....	3,700,000
Preferred.....	1,300,000
Burlington, Cedar Rapids & Northern.....	5,500,000
Buffalo, Rochester & Pittsburg.....	6,000,000
Preferred.....	6,000,000

	Amount.
Canada Southern	15,000,000
Canadian Pacific	65,000,000
Central of New Jersey	18,563,200
Central Iowa	9,100,000
First preferred	907,000
Second preferred	1,167,800
Central Pacific	68,000,000
Charlotte, Columbia & Augusta	2,578,000
Chesapeake & Ohio	15,906,138
First preferred	8,447,800
Second preferred	11,594,000
Chicago & Alton	14,091,000
Preferred	3,479,500
Chicago & Northwestern	41,373,000
Preferred	22,325,200
Chicago, St. Paul, Minneapolis & Omaha	21,403,293
Preferred	12,646,833
Chicago, Rock Island & Pacific	50,000,000
Chicago, Burlington & Quincy	76,385,700
Chicago, Milwaukee & St. Paul	39,680,361
Preferred	21,555,900
Chicago & Eastern Illinois	3,000,000
Chicago, St. Louis & Pittsburg	10,000,000
Preferred	20,000,000
Chicago & Indiana Coal Railway Company	2,197,800
Preferred	1,465,200
Cincinnati, New Orleans & Texas Pacific	3,000,000
Cincinnati, Indianapolis, St. Louis & Chicago	10,000,000
Cincinnati, Jackson & Mackinac	8,320,000
Preferred	4,680,000
Cleveland & Pittsburg Guaranteed	11,243,736
Cleveland, Columbus, Cincinnati & Ind's	14,991,800
Columbia & Greenville	1,000,000

	Amount.
Columbia & Greenville preferred	1,000,000
Columbus, Hocking Valley & Toledo	11,700,000
Delaware, Lackawanna & Western	26,200,000
Morris & Essex	15,000,000
New York, Lackawanna & Western	10,000,000
Dubuque & Sioux City	5,000,000
Denver & Rio Grande	38,000,000
Preferred	23,650,000
Denver & Rio Grande Western	7,500,000
Denver, South Park & Pacific	3,500,000
Des Moines & Fort Dodge	4,283,100
Preferred	763,000
Detroit, Mackinac & Marquette	4,750,000
Detroit, Bay City & Alpena	1,670,000
East Tennessee, Virginia & Georgia	27,500,000
First preferred	11,000,000
Second preferred	18,500,000
Elizabethtown, Lexington & Big Sandy	5,000,000
Evansville & Terre Haute	3,000,000
Flint & Pere Marquette preferred	6,500,000
Green Bay, Winona & St. Paul	8,000,000
Preferred	2,000,000
Harlem	8,518,100
Preferred	1,381,500
Houston & Texas Central	10,000,000
Illinois Central	40,000,000
Leased line, 4 per cent stock	10,000,000
Indiana, Bloomington & Western: Associated first installment paid; Associated full assessment paid	10,000,000
Joliet & Chicago	1,500,000
Kentucky Central	6,600,000
Keokuk & Western	4,000,000

	Amount.
Kingston & Pembroke	4,500,000
Lake Erie & Western	11,840,000
Preferred	11,840,000
Lake Shore & Michigan Southern	49,466,500
Long Island	10,000,000
Louisville & Nashville	30,000,000
Louisville, New Albany & Chicago	5,000,000
Manhattan Railroad Consolidated	23,895,630
Marquette, Houghton & Ontonagon	2,378,600
Preferred	3,278,500
Mexican Central (Limited)	35,000,000
Milwaukee, Lake Shore & Western	2,000,000
Preferred	5,000,000
Milwaukee & Northern	4,131,000
Michigan Central	18,738,204
Missouri Pacific	45,000,000
Missouri, Kansas & Texas	46,405,000
Mobile & Ohio Railroad Associated	5,320,600
Morgan's Louisiana & Texas R. R. & S. S.	1,004,100
Minneapolis & St. Louis	6,000,000
Preferred	4,000,000
Minneapolis, Sault Ste. Marie & Atlantic	2,426,000
Preferred	2,426,000
New York Central & Hudson River	89,428,300
New York, New Haven & Hartford	15,500,000
Boston & N. Y. Air Line preferred 4 pc	3,000,000
New York, Lake Erie & Western	78,000,000
Preferred	8,536,900
New York, Ontario & Western	58,113,982
New York & New England	20,000,000
New Jersey & New York	1,500,000
Preferred	800,000

	Amount.
New York, Chicago & St. Louis	28,000,000
Assented	
Preferred	22,000,000
Assented	
New York, Susquehanna & Western	13,000,000
Preferred	8,000,000
Northern Pacific	49,000,000
Preferred	37,936,776
Nashville, Chattanooga & St. Louis	6,688,375
Norfolk & Western	7,000,000
Preferred	22,000,000
Norfolk Southern	1,000,000
Ohio & Mississippi	20,000,000
Preferred	4,030,000
Ohio Southern	3,840,000
Omaha & St. Louis preferred	2,220,500
Oregon & California	7,000,000
Preferred	12,000,000
Oregon & Transcontinental Company	40,000,000
Oregon Short Line	15,265,000
Oregon Improvement Company	7,000,000
Oregon Railway & Navigation Company	24,000,000
Philadelphia & Reading, 1st assessment paid	
Second assessment paid	34,702,000
Third assessment paid	
All assessments paid	
Preferred first assessment paid	
Second assessment paid	1,286,800
Third assessment paid	
All assessments paid	
Pittsburg, Fort Wayne & Chicago Guar'd	19,714,285
Special	10,776,600
Pitts., McK'pt & Youghiogheny Consol. Stk.	3,000,000

	Amount.
Peoria, Decatur & Evansville	8,400,000
Richmond & Alleghany Reorganization Cert. Stamped assessment paid	5,000,000
Richmond & Danville	5,000,000
Richmond & W. P't. Railway & W. Co.	40,000,000
Preferred	5,000,000
Rome, Watertown & Ogdensburg	5,293,900
Utica & Black River Guaranteed	2,223,000
South Carolina	4,204,160
Southern Pacific Company	88,076,200
St. Louis, Alton & Terre Haute	2,300,000
Preferred	2,468,400
Belleville & Southern Illinois preferred	1,275,000
St. Louis & San Francisco	11,954,300
Preferred	10,000,000
First preferred	4,500,000
St. Louis, Arkansas & Texas	9,555,000
St. Paul & Duluth	4,055,400
Preferred	5,377,003
St. Joseph & Grand Island	4,500,000
St. Paul, Minneapolis & Manitoba	20,000,000
Texas & Pacific Trust c'tf's, all ass'ts paid	32,188,700
Toledo & Ohio Central	1,592,000
Preferred	3,108,000
United New Jersey Railroad & Cons. Co	21,240,400
Union Pacific	60,868,500
Utah Central	4,250,000
Virginia Midland	6,000,000
Wabash, St. Louis & Pacific; Full-paid P. C. certificates	28,419,500
Full-paid preferred; Full-paid P. C. certificates	24,223,200
Wheeling & Lake Erie	3,600,000

Miscellaneous Stocks.

	Amount.
Bankers' & Merchants' Telegraph	3,000,000
Boston Land Company	800,000
Canton Company, Baltimore	4,500,000
Chartiers Valley Gas Company	3,000,000
Central New Jersey Land Improvement	2,200,000
Consolidated Gas Company	35,430,000
Delaware & Hudson Canal	24,500,000
Equitable Gas-Light Company	3,000,000
Iron Steamboat Company	2,000,000
Manhattan Beach Company	5,000,000
Philadelphia Company	7,500,000
Pullman's Palace Car Company	15,927,200
Southern & Atlantic Telegraph	948,875
Sutro Tunnel Company	20,000,000
Western Union Telegraph	81,200,000
Northwestern Telegraph	2,500,000
Central & South American Telegraph	4,006,600
Commercial Telegraph Company	1,800,000
Preferred	200,000
Mexican Telegraph Company	1,500,000
Joliet Steel Company	2,666,000

Coal and Mining Stocks.

American Coal	1,500,000
Consolidation Coal of Maryland	10,250,000
Cumberland Coal & Iron	500,000
Colorado Coal & Iron Company	10,000,000
Cameron Iron & Coal Company	2,720,900
Columbus & Hocking Coal & Iron	4,700,000
Marshall Consolidated Coal Company	2,000,000
Maryland Coal Company	4,400,000
New York & Perry Coal & Iron Company	3,000,000

	Amount.
New Central Coal Company	5,000,000
Pennsylvania Coal	5,000,000
Quicksilver Mining Company	5,708,700
Preferred	4,291,300
Tennessee Coal, Iron & Railroad Company	10,000,000

Express Stocks.

Adams Express	12,000,000
American Express	18,000,000
United States Express	10,000,000
Wells Fargo Express	6,250,000
Pacific Mail Steamship Company	20,000,000

Railroad Bonds.

Atchison, Topeka & Santa Fé 4½'s....1920,	4,687,000
Sinking-fund 6's....1911,	12,348,000
Atlantic & Pacific Guar. 1st gold 4's..1937,	17,610,000
Beech Creek first gold 4's....1936,	5,000,000
Baltimore & Ohio first 6's, Park'g bch, 1919,	3,000,000
5's gold....1885–1925, } Registered	10,000,000
Boston, Hoosac Tunnel & Wn. Deb. 5's, 1913,	2,000,000
Burlington, C. R. & Northern 1st 5's..1906,	6,500,000
Con. 1st & Col. Tr., gold 5's....1934, } Registered	5,000,000
Minneapolis & St. Louis 1st g 7's g'd, 1927,	150,000
Iowa City & Western 1st gold 7's...1909,	456,000
Cedar Rapids, Iowa, F's & N. 1st 6's, 1920,	825,000
1st 5's....1921,	1,905,000
Buffalo, N. Y. & Ph. Ry. Con. 1st 6's, 1921, } Trust certificates	11,000,000
Railroad general 6's....1924, } Trust certificates	3,700,000

		Amount.
Canada Southern 1st in't gt'd 5's	1908,	14,000,000
2d mortgage 5's	1913,	6,000,000
Registered		
Central Iowa 1st mortgage 7's	1899,	3,700,000
Coupons off		
Eastern division 1st 6's	1912,	1,515,000
Illinois division 1st 6's	1912,	1,520,000
Central R. R. & Bkg. Co. G. col. g 5's,	1937,	5,000,000
Chesapeake & Ohio Pur. M'y. Fd.	1898,	2,300,000
6's gold, Series A	1908,	2,000,000
6's gold, Series B	1908,	15,000,000
Coupons off		
Small bonds	1908,	
Coupons off		
Extension coupons, gold 4's	1986,	
Regular 4's	1986,	
6's, currency	1918,	10,122,500
Small bonds	1918,	
Mortgage 6's	1911,	2,000,000
Chesapeake, Ohio & S. W. Mtge. 5–6's,	1911,	6,676,000
2d mortgage 6's	1911,	2,495,000
Chicago & Alton 1st mortgage 7's	1893,	2,383,000
Sinking fund, gold 6's	1903,	2,655,000
Louisiana & Missouri River 1st 7's	1900,	1,785,000
2d 7's	1900,	300,000
St. Louis, Jacksonville & C. 1st 7's	1894,	2,365,000
1st guaranteed (564) 7's	1894,	564,000
2d mortgage (360) 7's	1898,	44,000
2d guaranteed (188) 7's	1898,	188,000
Mississippi River B'ge 1st S. F. g 6's,	1912,	660,000
Chicago, B. & Q. consolidated 7's	1903,	30,000,000
5's, sinking fund	1901,	2,500,000
5's, debentures	1913,	9,000,000

		Amount.
C. B. & Q., Iowa Division S. F. 5's	1919,	3,000,000
4's	1919,	10,591,000
Denver Division 4's	1922,	7,968,000
4's	1921,	4,300,000
Nebraska Extension 4's	1927,	7,600,000
Registered		400,000
Chicago, Burlington & North. 1st 5's	1926,	9,000,000
Debenture 6's	1896,	2,250,000
Chicago, R. I. & Pacific 6's, coupon	1917,	12,500,000
6's, registered	1917,	
Extension & Col. 5's	1934,	13,960,000
Registered		
Des Moines & Fort Dodge 1st 4's	1905,	1,200,000
1st 2½'s	1905,	1,200,000
Extension 4's		672,000
Keokuk & Des Moines 1st Mg. 5's	1923,	2,750,000
Central Railroad of N. J., 1st 7's	1890,	5,000,000
1st consolidated 7's	1899,	25,000,000
Assented		
Convertible 7's	1902,	5,000,000
Assented		
Convertible debenture 6's	1908,	5,000,000
Interim bond certificates		12,000,000
Lehigh & W. B. Con. guaranteed	1900,	11,500,000
Assented		
Am. Dock & Improvement Co. 5's	1921,	5,000,000
M. & St. Paul's 1st M. 8's, P. D.	1898,	3,674,000
2d 7 3–10 P. D.	1898,	1,241,000
1st 7's $ gold, R. D.	1902,	3,804,000
1st 7's £ gold, R. D.	1902,	
1st mortgage La. Consolidated 7's,	1893,	5,264,000
1st mortgage I. & M. 7's	1897,	3,198,000
1st mortgage I. & D. 7's	1899,	541,000

	Amount.
M. & St. P. 1st mortgage C. & M. 7's.1903,	2,393,000
Consolidated 7's.....1905,	35,000,000
1st mortgage 7's, I. & D. extension, 1908,	3,505,000
1st 6's, Southwestern Division....1909,	4,000,000
1st 5's Louisiana C. & Dav.....1919,	3,000,000
1st South Min. Division 6's.....1910,	7,432,000
1st H. & D. Division 7's.....1910,	5,680,000
5's.....1910,	585,000
Chicago & Pacific Division 6's....1910,	2,500,000
1st Chicago & Pacific W. 5's,.....1921,	24,540,000
Chicago & Mo. River Division 5's.1926,	2,049,000
Mineral Point Division 5's.....1910,	2,840,000
C. & Lake Superior Division 5's...1921,	1,360,000
Wisconsin & Min. Division 5's....1921,	4,755,000
Terminal 5's.....1914,	4,666,000
Far. & So. 6's Assu.....1924,	1,250,000
Inc. convertible sinking fund 5's..1916,	2,000,000
Dakota & Great Southern 5's.....1916,	1,000,000
Chicago and Northwestern Con. 7's, 1915,	12,900,000
Coupon gold 7's.....1902, } Registered gold 7's.....1902,	48,000,000
Sinking fund 6's.....1879–1929, } Registered.....	6,305,000
5's.....1879–1929, } Registered.....	8,155,000
Debenture 5's.....1933, } Registered.....	10,000,000
25 years debenture 5's.....1909, } Registered.....	4,000,000
Extension gold 4's.....1886–1926,	8,190,000
Escanaba & Lake Superior 1st 6's...1901,	720,000
Des Moines & Minneapolis 1st 7's...1907,	600,000
Iowa Midland 1st mortgage 8's.....1900,	1,350,000

	Amount.
Peninsula 1st convertible 7's.....1898,	152,000
Chicago & Mil. 1st mortgage 7's....1898,	1,700,000
Winona & St. Pet's 2d 7's.....1907,	1,592,000
Milwaukee & Madison 1st 6's.....1905,	1,600,000
Ottumwa C. F. & St. P. 1st 5's.....1909,	1,600,000
Northern Illinois 1st 5's.....1910,	1,500,000
C. C. C. & Ind's 1st 7's sinking fund...1899,	3,000,000
Consolidated mortgage 7's.....1914, } Sinking fund 7's.....1914,	7,500,000
General consolidated gold 6's.....1934, } Registered.....	3,500,000
Chicago, St. Paul, Min. & O'a Con. 6's, 1930,	22,839,000
Chicago, St. Paul & Min. 1st 6's.....1918,	3,000,000
N. Wisconsin 1st mortgage 6's.....1930,	800,000
St. Paul & Sioux City 1st g. 6's.....1919,	6,080,200
Chicago & Eastern Illinois 1st S. F. C'y, 1907,	3,000,000
Small bonds.....	
1st consolidated 6's gold.....1934,	3,000,000
Chicago, St. L. & Pitts. 1st con. g. 5's, 1932, } Registered.....	22,000,000
Chicago & Western Ind. 1st S. F. g. 6's, 1919,	2,500,000
General mortgage g. 6's.....1932,	8,896,666
Chicago & St. Louis 1st 6's.....1915,	1,500,000
Chicago & Ind. Coal Railway 1st 5's...1936,	3,689,000
Cin., Ind., St. Louis & Chicago 1st g. 4's, 1936, } Registered.....	1,255,000
Cin., Jack. & Mac. 1st con. g. 5's.....1936,	1,400,000
Columbia & Greenville 1st 6's.....1916,	2,000,000
2d 6's.....1926,	1,000,000
Columbus, Hocking V. & Tol. Con. g. 5's, 1931,	14,500,000
General mortgage gold 6's.....1904,	2,000,000
Columbus & Cincinnati Midland 1st 6's, 1914,	2,000,000
Delaware, Lack. & Western Conv. 7's..1892,	600,000

		Amount.
Del. Lack. & W. Mortgage 7's	1907,	10,000,000
Syracuse, Binghamton & N. Y. 1st 7's	1906,	1,750,000
Morris & Essex 1st mortgage 7's	1914,	5,000,000
2d 7's	1891,	3,000,000
Bonds 7's	1900,	281,000
7's of	1871–1901,	4,991,000
1st consolidated guaranteed 7's	1915,	25,000,000
New York, Lack. & Western 1st 6's	1921,	12,000,000
Construction 5's	1923,	5,000,000
Del. & Hudson Canal 1st reg. 7's	1891,	4,988,000
1st extension regular 7's	1891,	549,000
Coupon 7's	1894,	4,829,000
Registered 7's	1894,	
1st Pennsylvania Division c. 7's	1917,	10,000,000
Registered	1917,	
Albany & Susquehanna 1st 7's	1888,	1,000,000
1st con. guaranteed 7's	1906,	3,000,000
Registered		
6's	1906,	5,488,000
Registered		
Rens'r & Saratoga 1st coupon 7's	1921,	2,000,000
1st regular 7's	1921,	
Denver & Rio Grande 1st Con. g. 4's	1936,	22,575,000
Denver & Rio Grande 1st mort. g. 7's	1900,	6,382,500
Denver, S. P'k & Pacific 1st g. 7's	1905,	1,800,000
Denver & Rio Grande W'n 1st g. 6's	1911,	5,857,000
Assented		
Detroit, Mackinac & Marquette 1st 6's	1921,	2,280,000
Land grant 3½ S. A.	1911,	4,560,000
Detroit, Bay City & Alpena 1st g. 6's	1913,	2,300,000
E. Tenn., Virginia & Georgia 1st 7's	1900,	3,500,000
Divisional 5's	1930,	3,106,000
E. Tenn., Va. & Ga. Ry. con. 1st g. 5's	1956,	12,770,000

	Amount.
E. & W. of Alabama 1st con. gold 6's..1926,	1,709,000
Elizabeth City & Norfolk S. F. deb. cert. 6's..	250,000
1st mortgage g. 6's.....1920,	900,000
Elizabethtown, Lex. & Big Sandy 6's..1902,	3,500,000
Erie 1st mortgage exten'd g. 7's.....1897,	2,482,000
2d exten'd g. 5's.....1919,	2,149,000
3d exten. g. 4½'s.....1923,	4,618,000
4th exten. g. 5's.....1920,	2,926,000
5th 7's.....1888,	709,500
1st con. gold 7's.....1920,	16,890,000
1st con. gold F'd 7's.....1920,	3,705,997
Reorganization 1st lien 6's.....1908,	2,500,000
Long Dock bonds 7's.....1893,	3,000,000
Consolidated gold 6's.....1935,	4,500,000
Buffalo, New York & Erie 1st 7's...1916,	2,380,000
N. Y. L. E. & W. new 2d con. 6's...1969,	33,597,400
Collateral Trust 6's.....1922,	5,000,000
Funding coupon 5's.....1885–1969,	4,032,000
Buffalo & Southwestern mort. g. 6's, 1908, Small.....	1,500,000
Evansville & Terre H. 1st con. g. 6's..1921,	3,000,000
Mount Vernon 1st 6's.....1923,	375,000
Evansville & In's 1st con. gtd g. 6's...1926,	1,020,000
Eureka Springs Railway 1st 6's g.....1933,	500,000
Flint & Pere Marquette mort. g. 6's...1920,	5,000,000
Fort Worth & Denver City 1st 6's.....1921,	6,448,000
Galveston, Har. & San Ant. 1st g. 6's..1910,	4,800,000
2d mortgage g. 7's.....1905,	1,000,000
Western Division 1st g. 5's.....1931,	13,500,000
2d 6's.....1931,	6,750,000
Grand Rapids & Ind. general 5's.....1924, Registered.....	3,217,000
Green Bay, W. & St. Paul 1st 6's.....1911,	1,600,000

	Amount.
Gulf, Colorado & Santa Fé 1st 7's.....1909,	11,724,000
Gold 6's.....1923,	6,000,000
Hannibal & St. Joseph consolidated 6's, 1911,	6,000,000
Henderson Bridge Company 1st g. 6's, 1931,	2,000,000
Houston & Texas Cent. 1st M'n L. 7's, 1891,	6,896,000
1st Western Division 7's.....1891,	2,375,000
1st Waco & N. 7's.....1903,	1,140,000
2d C. Main L. 8's.....1912,	4,118,000
General mortgage g. 6's.....1921, Trust Company receipts.....	4,325,000
Houston, E. & W. Texas 1st g. 7's.....1898,	1,344,000
Illinois Central.	
1st gold 4's.....1951, Registered.....	1,500,000
1st gold 3½'s.....1951, Registered.....	2,500,000
Springfield Division coupon 6's.....1898,	1,600,000
Middle Division regular 5's.....1921,	600,000
Chicago, St. L. & N. O. Ten. lien 7's, 1897,	541,000
1st con. 7's.....1897,	857,000
2d mortgage 6's.....1907,	80,000
Gold 5's.....1951, Registered.....	18,000,000
Dubuque & Sioux City 2d div. 7's...1894,	586,000
Cedar Falls & Minneapolis 1st 7's...1907,	1,334,000
Indiana, Bl'n & W. 1st preferred 7's...1900,	1,000,000
1st 5–6's trust receipts.....	3,408,000
2d 5–6's trust receipts.....	1,477,000
Eastern Division trust receipts.....	2,950,000
Ind's, D. & Sp. 1st 7's Ex. F. coupon..1906,	1,613,000
International & Great N'n 1st 6's gold, 1919,	7,954,000
Coupon 6's.....1909,	7,054,000
Kentucky Central Railway gold 4's...1987,	6,600,000

	Amount.
Knoxville & Ohio 1st 6's gold.....1925,	2,000,000
Lake Erie & Western 1st gold 5's.....1937,	5,920,000
Lake Shore & Michigan Southern.	
Cleveland, Pain'le & Ash. 7's.....1892,	920,000
Buffalo & Erie new bonds 7's.....1898,	2,784,000
Kalamazoo & W. Pig'n 1st 7's.....1890,	400,000
Detroit, Montreal & Toledo 1st 7's..1906,	924,000
Lake Shore Division bonds 7's.....1899,	1,356,000
Consolidated coupon 1st 7's.....1900, } Registered 1st.....1900, }	25,000,000
Coupon 2d 7's. .....1903, } Registered 2d.....1903, }	25,000,000
Mahoning Coal Railroad 1st 5's.....1934,	1,500,000
Long Island 1st mortgage 7's.....1898,	1,500,000
1st consolidated g. 5's.....1931,	5,000,000
New York & Manhattan B'h 1st 7's, 1897,	500,000
New Y., B. & Man. B. 1st con. g. 5's, 1935,	783,000
Louisville & Nashville consolidated 7's, 1898,	7,070,000
Cecilian Branch 7's.....1907,	1,000,000
New Orleans & Mobile 1st g. 6's....1930,	5,000,000
2d 6's.....1930,	1,000,000
E. H. & Nashville 1st g. 6's.....1919,	2,400,000
General mortgage g. 6's.....1930,	20,000,000
Pensacola Division 6's.....1920,	600,000
St. Louis Division 1st g. 6's.....1921,	3,500,000
2d g. 3's.....1980,	3,000,000
Nashville & Dec. 1st 7's.....1900,	1,900,000
South & N. Ala. sinking fund 6's...1910,	2,000,000
Louisville, Cin. & Lexington g. 6's..1931,	7,000,000
Trust bonds g. 6's.....1922,	10,000,000
Ten-forty g. 6's.....1924,	5,000,000
5 per cent fifty year gold bonds.....1937,	1,350,000
Pens. & At. 1st 6's gold guaranteed, 1921,	3,000,000

	Amount.
Louisville, N. Albany & Chicago 1st 6's, 1910,	3,000,000
Consolidated gold 6's.....1916,	3,500,000
Louisville, N. O. & Texas 1st gold 4's, 1934,	11,140,000
2d mortgage 5's.....1934,	8,117,000
Memphis & Charleston 6's gold.....1924,	1,000,000
Metropolitan Elevated 1st g. 6's.....1908,	10,818,000
2d 6's.....1899,	4,000,000
Mexican Central 1st mortgage 7's.....1911, Ex-coupon 6–7–8..... New assented 4's.....	41,170,000
Income bonds..... 1911,	8,734,000
Michigan Central 1st consolidated 7's.1902,	8,000,000
5's.....1902,	2,000,000
6's.....1909,	1,500,000
Coupon 5's..... .....1931, Registered 5's.....1931,	4,000,000
Jack., Lan. & Sag. 6's.....1891,	1,100,000
Milwaukee & Nor. 1st main line, 6's...1910,	2,155,000
On extension 1st 6's.....1913,	1,976,000
Milwaukee, Lake Shore & W'n 1st g. 6's, 1921,	4,350,000
Convertible debenture 5's.....1907,	600,000
Michigan Division 1st 6's.....1924,	1,281,000
Ashland Division 1st g. 6's.....1925,	1,000,000
Minneapolis & St. Louis 1st g. 7's.....1927,	950,000
Iowa extension 1st g. 7's.....1909,	1,015,000
2d mortgage 7's.....1891,	500,000
Southwestern extension 1st g. 7's...1910,	636,000
Pacific extension 1st g. 6's.....1921,	1,382,000
Improvement & Equipment 6's.....1922,	2,000,000
Minneapolis & Pacific 1st mortgage 5's, 1936,	3,035,000
Minneapolis & Northwestern 1st 5's g., 1934,	7,783,000
Minn., Sault Ste. Marie & Atl. 1st g. 5's, 1926,	4,000,000
Missouri, Kansas & T. Gen. Cons. g. 6's, 1920,	35,815,000

		Amount.
M. K. & T. Gold 5's	1920,	9,284,000
Consolidated g. 7's	1904–5–6,	14,877,000
2d mortgage Inc	1911,	630,000
H. & Central Missouri 1st g. 7's	1890,	664,000
Mobile & Ohio new mortgage g. 6's	1927,	7,000,000
Collateral trust 6's	1892,	59,000
1st extension 6's	1927,	1,000,000
St. Louis & Cairo g. 4's guaranteed,	1931,	4,000,000
Morgan's Louisiana & Texas 1st g. 6's,	1920,	1,494,000
1st 7's	1918,	5,000,000
Nashville, Chat. & St. Louis 1st 7's	1913,	6,800,000
2d 6's	1901,	1,000,000
New York Central 6's	1887,	2,391,000
Debt. certificates extended 5's	1893,	6,450,000
New York & Hudson 1st coupon 7's,	1903,	30,000,000
1st registered	1903,	
Debenture 5's	1904,	7,850,000
Registered		
Harlem 1st mortgage 7's, coupon	1900,	12,000,000
7's, registered	1900,	
New Jersey Junction guar. 1st 4's	1986,	2,000,000
Registered certificates		
New York Elevated 1st mortgage 7's	1906,	8,500,000
New York, Penn. & Ohio prior lien 6's,	1895,	8,000,000
New York City & Northern Gen. M. 6's,	1910,	4,000,000
Trust Company receipts		
Assented		
New York & New England 1st 7's	1905,	6,000,000
1st 6's	1905,	4,000,000
N. Y., Ch. & St. L. 1st 6's T't Rec. Ass'd		15,000,000
New Trust Company receipts		
2d 6's	1923,	10,000,000
New York, Ontario & W. 1st gold 6's	1914,	3,000,000

	Amount.
New York, Susq'a & W'n debent. 6's..1897, Coupons off.....	600,000
1st refunding 5's..... 1937,	3,750,000
2d mortgage 4½'s.....1937,	636,000
Midland Railroad of New J. 1st 6's, 1910,	3,500,000
N. Y., N. Haven & Hartford 1st reg. 4's, 1903,	2,000,000
N. Y., Texas & Mexican guar. 1st 4's..1912,	1,442,500
Northern Pacific Gen. 1st M. R. R., coupon L'd Gt. gold 6's, 1921, regular.....	53,309,000
Gen. 2d M. R. R., coupon..... L. G. sinking fund gold 6's, 1933, regular	20,000,000
Dividend scrip..... Extended.....	4,640,821
James River Valley 1st 6's, gold....1936,	963,000
Spokane & Pal. 1st sinking fund g. 6's, 1936,	688,000
St. Paul & N. Pacific Gen. g. 6's....1923, Registered certificates.....	6,300,000
Helena & Red M'n 1st g. 6's.....1937,	400,000
Duluth & Manitoba 1st g. 6's.....1936,	1,650,000
Hel. B. Valley & Butte 1st 6's, g....1937,	600,000
Nor. Pacific Term'i Co. 1st gold 6's...1933,	3,000,000
New Orleans Pacific 1st 6's, gold..... 1920, Coupons off..... Trust Company receipts.....	6,720,000
N. Orleans & N. E'n prior lien gold 6's, 1915,	1,050,000
Norfolk & Western Gen'l Mort. 6's....1931,	6,902,000
New River 1st 6's.....1932,	2,000,000
Improvement & extension g. 6's.....1934,	3,500,000
Adjustment mortgage g. 7's.....1924,	1,500,000
Ogdensburg & L. Champlain 1st con. 6's, 1920,	3,500,000
Ohio & Miss. Cons. sinking fund 7's...1898,	3,435,000
Consolidated 7's.....1898,	3,066,000
2d consolidated 7's.....1911,	3,715,000

	Amount.
Ohio & Miss. 1st Springfield Division 7's. 1905,	3,000,000
1st general 5's. 1932,	3,216,000
Ohio Central 1st Ter'l Trust 6's. 1920,	600,000
1st Min'l Division 6's. 1921,	300,000
Ohio River Railroad 1st g. 5's. 1936,	2,000,000
Ohio Southern 1st mortgage 6's. 1921,	2,100,000
Omaha & St. Louis Railway 1st 4's. . . . 1937,	2,717,000
Oregon & California 1st g. 6's. 1921,	9,000,000
Oregon & Transcontinental 6's. . . 1882–1922,	10,063,000
Oregon Improvement Co. 1st g. 6's. . . . 1910,	5,000,000
Oregon R'y & Navigation 1st g. 6's. . . . 1909,	6,000,000
Consolidated mortgage g. 5's. 1925,	9,137,000
Panama sinking fund Sub'y g. 6's. 1910,	2,747,000
Peoria, Decatur & Evansville 1st 6's. . . 1920,	1,287,000
Evansville Division 1st 6's. 1920,	1,470,000
2d mortgage g. 5's. 1927,	2,088,000
Peoria & Pekin Union 1st g. 6's. 1921,	1,500,000
2d mortgage g. 4½'s. 1921,	1,499,000
Central Pacific, gold bonds 6's. 1895,	
" " 1896,	25,883,000
" " 1897,	
" " 1898,	
San Joaq'n Branch g. 6's. 1900,	6,080,000
California & Oregon 1st g. 6's. 1888,	6,000,000
Series B. g. 6's. 1892,	5,860,000
Land grant g. 6's. 1890,	9,436,000
Mortgage bond 6's. 1936,	12,000,000
Western Pacific bonds 6's. 1899,	2,735,000
Nor. R'y (Cal.) 1st g. 6's guaranteed. 1907,	3,964,000
Southern Pacific of Cal. 1st g. 6's, 1905–'12,	38,447,000
S. Pacific of Arizona guar. 1st 6's, 1909–'10,	10,000,000
Southern Pacific of N. Mexico 1st 6's, 1911,	5,000,000

		Amount.
Union Pacific 1st 6's	1896,	27,229,000
" "	1897,	
" "	1898,	
" "	1899,	
Land grants 7's	1888–'89,	1,270,000
Sinking fund 8's	1893,	14,348,000
Registered 8's	1893,	
Collateral trust 6's	1908,	4,423,000
5's	1907,	5,583,000
Kansas Pacific 1st 6's	1895,	2,240,000
1st 6's	1896,	4,063,000
Denver Division 6's assented	1899,	6,242,000
1st consolidated 6's	1919,	13,855,000
Central Branch U. P. F'd coup. 7's	1895,	630,000
Atchison, Col. & Pacific 1st 6's	1905,	3,672,000
Atchison Jew'l Co. & W. 1st 6's	1905,	542,000
Oregon Short Line 1st 6's	1922,	14,931,000
Utah Southern Gen'l mortgage 7's	1909,	1,950,000
Extension 1st 7's	1909,	1,950,000
Missouri Pacific 1st consolidated 6's	1920,	20,184,000
3d mortgage 7's	1906,	3,328,000
Pacific Railway of Mo. 1st mort. 6's	1888,	7,000,000
2d mortgage 7's	1891,	2,573,000
Verdig's Valley, Ind. & W. 1st 5's	1926,	750,000
Leroy & City Val. Air Line 1st 5's	1926,	520,000
St. L. & San Fran. 2d 6's, class A	1906,	500,000
6's, class C	1906,	2,400,000
6's, class B	1906,	2,766,500
1st 6's Pierce C. & O. branch		1,090,000
Equipment 7's	1895,	650,000
General mortgage 6's	1931,	7,732,000
5's	1931,	5,000,000
South Pacific R. of Missouri 1st 6's	1888,	7,144,500

	Amount.
Kansas City & S. W'n 1st 6's gold...1916,	744,000
Ft. S'th & Van B. Bdg. 1st 6's.....1910,	475,000
St. Louis, Kan. & S. W'n 1st 6's....1916,	735,000
Texas & Pacific Railway 1st 6's.....1905, Ex. coupon.....	3,784,000
Consolidated 6's, trust receipts.....	9,316,000
Inc. Land Grant Asst'd trust receipts....	7,992,000
R. G. 6's 1930, trust receipts.....	13,028,000
General Mort. & T. trust receipts.....	2,859,000
Pennsylvania Railroad Company.	
Pennsylvania Co.'s guar. 4½ 1st coup., 1921, Regular.....1921,	15,000,000
Pittsburg, Chicago & St. Louis 1st coup. 7's.....1900,	2,706,000
1st regular 7's.....1900,	4,157,000
2d 7's.....1913,	2,500,000
Pitts., Fort Wayne & C. 1st 7's.....1912,	5,250,000
2d 7's.....1912,	5,160,000
3d 7's.....1912,	2,000,000
Clev. & Pittsburg Cons. S. F. 7's....1900,	2,292,000
4th sinking fund 6's.....1892,	1,105,000
St. L., V'a & T. H. 1st gtd 7's.....1897,	1,899,000
2d 7's.....1898,	1,000,000
2d guaranteed 7's.....1898,	1,600,000
Phil. & Reading Inc. M. Coupon 7's...1896, Trust receipts..... 4th assessment paid.....	10,000,000
Debenture coupon 6's.....1893, Trust receipts..... 4th assessment paid.....	670,500
Debenture convertible 7's.....1893, Trust receipts..... 4th assessment paid.....	10,395,900

	Amount.
Phil. & R. Preferred 1st series con. 5's 1922, Trust receipts..... 4th assessment paid.....	6,000,000
2d assessment 5's.....1933, Trust receipts..... 4th assessment paid.....	5,000,000
Pine Creek Railway 6's of..... 1932,	3,500,000
Pittsburg, Cleveland & Tol. 1st 6's....1922,	2,400,000
Pittsburg Junction 1st 6's.....1922,	1,440,000
Pittsburg, McKeesp't & Y. 1st 6's....1932,	2,250,000
Rome, Wat. & Ogdensburg 1st 7's.....1891,	1,021,500
Consolidated 1st extension 5's.....1922,	6,337,000
Rochester & Pittsburg 1st 6's.....1921,	1,300,000
Consolidated 1st 6's.....1922,	3,920,000
Richmond & Alleghany 1st 7's.....1920, Trust Company receipts..... Stamped.....	5,000,000
Richmond & Danville cons. g. 6's.....1915,	6,000,000
Debenture 6's.....1927, Extension coupon.....	4,000,000
Con. mortgage gold 5's.....1936,	1,500,000
Atlanta & Char. 1st preferred 7's....1897,	500,000
Atlanta & Charlotte Inc.....1900,	750,000
Richmond & W. P't Ter'l Trust 6's...1897,	8,500,000
San A. & Arans Pass 1st g. 6's...1885–1916,	1,750,000
1886–1926,	2,598,000
Scioto Valley 1st consolidated 7's.....1910, Coupons off.....	603,000
St. Joseph & G'd Island 1st 6's.....1925,	7,000,000
St. Louis & Iron Mountain 1st 7's.....1892,	4,000,000
2d 7's.....1897,	6,000,000
Arkansas Branch 1st 7's.....1895,	2,500,000
Cairo & Fulton 1st 7's.....1891,	7,555,000

		Amount.
Cairo, Arkansas & T. 1st 7's	1897,	1,450,000
Gen. con. railroad & land-grant 5's,	1931,	38,201,000
St. Louis, Alt'n & T. H'te 1st 7's	1894,	2,200,000
2d mortgage preferred 7's	1894,	2,800,000
2d mortgage inconvertible 7's	1894,	1,700,000
Belleville & S. Ills. Railway 1st 8's	1896,	1,041,000
Belleville & Carond't 1st 6's	1923,	485,000
St. Louis, Ark. & Tex. 1st certi. 6's	1936,	12,870,000
2d certificates 6's	1936,	11,804,000
St. Paul, Minn. & Manitoba 1st 7's	1909,	4,991,000
Small		
2d 6's	1909,	8,000,000
Dakota extension 6's	1910,	5,676,000
1st consolidated 6's	1933,	23,444,000
Registered		
Reduced to 4½'s		
Registered		
Minneapolis Union 1st 6's	1922,	2,150,000
St. Paul & Duluth 1st 5's	1931,	1,000,000
South Carolina Railway 1st 6's	1920,	5,000,000
2d 6's	1931,	1,500,000
Shenandoah Valley 1st 7's	1909,	2,270,000
Trust Company receipts		
General mortgage 6's	1921,	4,113,000
Trust receipts		
Sodus Bay & S. 1st 5's gold	1924,	500,000
Texas Central 1st sinking fund 7's	1909,	2,145,000
1st mortgage 7's	1911,	1,254,000
Texas & New Orleans 1st 7's	1905,	1,620,000
Sabine Division 1st 6's	1912,	2,075,000
Tol. & Ohio Central 1st gold 5's	1935,	3,000,000
Tol., Peoria & Western 1st 7's	1917,	4,500,000
Trust Company receipts		

	Amount.
Tol., Ann A. & N. Michigan 1st 6's...1924,	2,120,000
Tol., Ann Arbor & G. T. 1st 6's gold..1921,	1,260,000
Tol., St. L. & Kansas City 1st g. 6's...1916,	2,000,000
Val'y Railway Co. of O. Con. gold 6's..1921,	1,700,000
Virginia Midland mortgage Inc. 6's...1927,	604,000
General mortgage 5's.....1936,	3,717,000
Wabash, St. L. & Pac. gen'l mort. 6's, 1920, Trust Company receipts.....	16,000,000
Chicago Division 5's.....1910,	4,500,000
Havana Division 6's.....1910,	1,600,000
Indianapolis Division 6's.....1921,	2,275,000
Detroit Division 6's.....1921,	2,052,000
Cairo Division 5's.....1931,	3,857,000
Wabash Railroad mortgage 7's..1879–1909,	2,000,000
Tol. & Wabash 1st extended 7's.....1890,	3,400,000
1st St. Louis Division 7's.....1889,	2,700,000
2d mortgage extended 7's.....1893,	2,500,000
Equip. bonds 7's.....1883,	600,000
Consolidated convertible 7's.....1907,	2,600,000
Great Western 1st mortgage 7's.....1888,	2,500,000
2d mortgage 7's.....1893,	2,500,000
Quincy & Tol. 1st mortgage 7's.....1890,	500,000
Hannibal & Naples 1st 7's.....1909,	500,000
Ill. & So. Iowa 1st extension 6's.....1912,	300,000
St. L., K., N. R'l Est'e & R. 7's.....1895,	3,000,000
Clarinda branch 6's.....1919,	264,000
St. Chas. B'ge 1st 6's.....1908,	1,000,000
Northern Missouri 1st mortgage 7's, 1895,	6,000,000
Wab., St. L. & P. Iowa trust receipts.....	2,269,000
West Shore 1st mortgage 4's guaranteed... Registered.....	50,000,000
Western Union coupon 7's.....1900, Registered.....	3,920,000

	Amount.
Northwestern Telegraph 7's.....1994,	1,250,000
Wheeling & Lake Erie 1st 5's.....1926,	3,000,000
Mutual Union Telegraph Sk'g F. 6's..1911,	5,000,000
Man. B. Improvement Co. limited 7's..1909,	1,000,000
Colorado, C'l & I'n 1st construction 6's, 1900,	3,500,000
Tenn. Coal, Iron & R. Con. 6's.....1901,	620,000
South Pittsburg 1st 6's.....1902,	720,000
Bir. Division 1st Con. 6's.....1917,	4,000,000
Col. & Hock'g Coal & Iron 6's g.....1917,	1,000,000

INCOME BONDS.

Atlantic & Pac. W'n Division Inc.....1910, Small.....	10,500,000
Central Division Inc.....1922,	2,100,000
Central Iowa coupon debenture certificates.	620,000
Chicago & Eastern Illinois income....1907,	1,000,000
Des M's & Fort D. 1st Inc. 6's.....1905,	1,200,000
Det., Mack. & Marquette Inc.....1921,	1,500,000
Elizabeth City & Nor. 2d Inc.....1970,	1,000,000
G. Bay, W. & St. Paul 2d Inc.....1911,	3,781,000
Ind., Bl'n & W. Cons. Inc. trust receipts...	4,560,000
Ind's, Decatur & Sp'd 2d Inc.....1906, Trust Company receipts.....	2,850,000
Lehigh & W. B're Coal Company.....1888, Small bonds.....1888,	1,119,200
Mil., Lake Shore & Western Income.....	500,000
Mobile & Ohio 1st preferred debenture.....	4,763,000
2d preferred debentures.....	1,850,000
3d preferred debentures.....	600,000
4th preferred debentures.....	900,000
New York, Lake Erie & W'n Inc. 6's..1977,	508,000
New York, Penn. & O. 1st Inc. acc. 7's, 1905,	35,000,000
Ohio Central Min. Division Inc. 7's...1921,	300,000

	Amount.
Ohio Southern 2d Income 6's.....1921,	2,100,000
Og'b'g & Lake Champlain Income....1920,	800,000
Small.....	200,000
Rochester & Pittsburg Income.....1921,	478,000
South Carolina Railway Income 6's...1931,	3,000,000
St. L., I. M. & S. 1st 7's preferred int. ac'e..	348,000
Sterling Iron & Railway Ser's B. Inc..1894,	418,000
Plain Income 6's.....1896,	491,000
Sterling Mountain Railway Income...1895,	476,000
St. L., Alton & T. H. Division bonds..1894,	1,357,000
St. Joseph & G'd Island 2d Income. ..1925,	1,680,000
Shenandoah Valley Income 6's.....1923,	2,500,000

PHILADELPHIA STOCK-EXCHANGE.

Railroad Stocks.

	Interest.
Atlantic & Pacific.....	100
Bell's Gap.....	50
Buffalo, New York & Philadelphia.....	50
Preferred.....	50
Common, assessment paid.....	50
Preferred, assessment paid.....	50
Camden & Atlantic.....	50
Preferred.....	50
Catawissa.....	50
1st preferred.....	50
2d preferred.....	50
Central of New Jersey.....	100
Clearfield & Jefferson.....	
Delaware & Bound Brook.....	100
Denver & Rio Grande.....	100

	Interest.
East Pennsylvania	50
Elmira & Williams	50
Preferred	50
Harrisburg	50
Huntingdon & Broad Top Mountain	50
Preferred	50
Lehigh Valley	50
Little Schuylkill	50
Minehill	50
Nesquehoning Valley	50
Norfolk & Western	100
Preferred	100
Norristown	50
Northern Central	50
Northern Pacific	100
Preferred	100
North Pennsylvania	50
Oregon & Transcontinental	100
Pennsylvania	50
Philadelphia & Erie	50
Philadelphia & Reading	50
Preferred	50
St. Paul & Duluth	100
Preferred	100
Sunbury & Lewis	50
Texas & Pacific	100
United Companies of New Jersey	100
West Jersey	50
West Jersey & Atlantic	50

Canal Stocks.

Chesapeake & Delaware	50
Lehigh Navigation	50

	Interest.
Morris	100
Preferred	100
Schuylkill Navigation, preferred	50
Common	50

Railroad Bonds.

Allegheny Valley, regular	7 3-10
Income	7
Baltimore & Ohio, E. Side	5
Belville Delaware, 1st	6
Consolidated	4
Bell's Gap, consolidated	6
Camden & Amboy, mortgage, coupon, 1889	6
Coupon, 1889	6
Catawissa, new	7
Camden & Atlantic, 1st	7
2d mortgage	6
Columbus & Cincinnati Midland	6
Connecting, 1st	6
Clearfield & Jefferson	6
Delaware Railroad, 1st	6
Delaware & Bound Brook, 1st	7
East Pennsylvania, 1st	7
Easton & Amboy	5
Elmira & Western, 1st, 1910	6
Perpetual	5
Huntingdon & Broad Top, 1st, gold	7
2d, 1895	7
3d, consolidated, 1895	5
Ithaca & Athens, 1st	7
Lehigh Valley, 1st, coupon	6
1st, regular, 1898	6
2d, regular, 1910	7

	Interest.
Lehigh Val. Consolidated mortgage, regular, 1923..	6
Consolidated mortgage, coupon, 1923.....	6
New Orleans Pacific, 1st, 1920.....	6
New York, Philadelphia & Norfolk, 1st.....	6
Income.....	6
Norfolk & Western, new railroad, 1st.....	6
Debenture.....	6
General mortgage.....	6
Northern Central, general mort., "A," coup., 1926..	5
General mortgage, series "B".....	5
General mortgage, coupon, 1904.....	6
Northern Pacific, general mortgage.....	6
2d, coupon or regular.....	6
North Pacific, 1st, 1896.....	7
General mortgage, coupon, 1903.....	7
General mortgage, regular, 1903.....	7
Debentures.....	6
Pennsylvania & New York C. & R., 1896.....	7
Regular and coupon, 1906.....	7
Pennsylvania, general, coupon, 1910.....	6
General, regular, 1910.....	6
Consolidated, regular, 1905.....	6
Consolidated, coupon, 1905.....	6
Consolidated, regular, 1919.....	5
Consolidated, coupon, 1919.....	5
Pennsylvania Company, regular, 1907.....	6
Regular, 1920.....	4½
Coupon, 1920.....	4½
Perkiomen, 1st, 1887.....	6
Philadelphia & Erie, 2d, 1888.....	7
General mortgage, 1920.....	5
Philadelphia & Reading, 1st series.....	5
Consolidated, 2d series.....	5

	Interest.
1st, 1910	6
2d, coupon, 1893	7
Improvement	6
Consolidated, coupon, 1911	7
Consolidated, regular, 1911	7
Consolidated, gold, 1911	6
General mortgage, gold, 1908	6
General mortgage, 1908	7
Philadelphia & Reading, income, coupon, 1896	7
Debenture	6
Scrip	6
Deferred income	6
Convertible adj. scrip, 1888	6
New convertible, 1893	7
Var. coupons on	
P. & R. C. & I., Var	7
Debenture	7
Philadelphia, Wilmington & Baltimore, trust certi.	4
Pittsburg, Cincinnati & St. Louis, coupon	7
Regular, 1900	7
Sham. Valley & Pott., coupon	7
Sham. Sun. & Lew	5
Shenandoah Valley, 1st	7
General mortgage	6
Steub. & Ind., 1st, coupon	5
Sunb. & Lewist., 1st	7
Sun., Haz. & W., 1928	5
2d, 1938	6
Texas & Pacific, 1st, gold, 1805	6
1st, R. G. Division, 1930	6
Consolidated, gold, 1905	6
Union & Titusville, 1st, 1890	7
United New Jersey, consolidated	6

	Interest.
United New Jersey, General mortgage	4
Warren & Farnsworth, 1st, 1896	7
West Chester, consolidated, 1891	7
West Jersey, 1st, 1896	6
1st, 1899	7
West Jersey & Atlantic, 1st, 1910	6
West Pennsylvania, 1st, 1893	6
Pittsburg Branch, coupon, 1896	6
West Shore, guaranteed	4

Canal Bonds.

Chesapeake & Delaware, extend	5
Lehigh Navigation, extend	4½
Convertible, gold, 1894	6
Railroad loan, 1897	6
Gold, 1897	6
Consolidated mortgage, 1911	7
General mortgage, 1924	4½
Greenwood Tract	7
Pennsylvania, mortgage, 1910	6
Schuylkill Navigation, 1st, 1897	6
2d, 1907	6
Boat loan	7
Boat loan	6
Susquehanna, 1918	6

BOSTON STOCK-EXCHANGE.

Railroad Stocks.

Atlantic & Pacific	100
Atchison, Topeka & Santa Fé	100
Boston & Albany	100

	Interest.
Boston, Concord & Montreal, preferred	100
Boston & Lowell	100
Boston & Maine	100
Boston & Providence	100
California Southern	100
Central Iowa	100
2d preferred	100
Central Massachusetts	100
Preferred	100
Cheshire, preferred	100
Chicago, Burlington & Quincy	100
Chicago, Burlington & Northern	100
Chicago & Eastern Illinois	100
Chicago & West Michigan	100
Cincinnati, Sand. & Cleveland	50
Cleveland & Canton	100
Preferred	100
Col., Spring. & Cincinnati	
Concord	50
Connecticut & Passumpsic	100
Connecticut River	100
Consolidated of Vermont, preferred	100
Detroit, Lansing & Northern	100
Preferred	100
Dayton & Ironton, preferred	
Eastern (Massachusetts)	100
Preferred	100
Eastern in New Hampshire	100
Fitchburg preferred	100
Flint & Pere Marquette	100
Preferred	100
Fort Scott & Gulf	100
Preferred	100

	Interest.
Iowa F. & Sioux City	100
Kansas City, Springfield & Memphis	100
Kansas City, Memphis & Birmingham	
Kansas City, Clinton & Springfield	100
Little Rock & Fort Smith	100
Louisville & Missouri River	100
Maine Central	100
Manchester & Lawrence	100
Marquette, Houghton & Ontonagon	100
Preferred	100
Mexican Central	100
Nashua & Lowell	100
New York & New England	100
Preferred	100
Northern, New Hampshire	100
Norwich & Worcester	100
Ogdensburg & Lake Champlain	100
Old Colony	100
Philadelphia, Wilmington & Baltimore	50
Portsmouth, Great Falls & Conway	100
Portland, Saco & Portsmouth	100
Pullman Palace Car	100
Revere Beach & Lynn	100
Rutland	100
Preferred	100
Summit Branch	50
Toledo, Cincinnati & St. Louis	50
Union Pacific	100
Vermont & Massachusetts	100
Wisconsin Central	100
Preferred	100
Worcester, Nashua & Rochester	100

Railroad Bonds.

	Interest.
Atchison, Topeka & Santa Fé, plain, 1920	5
Trust	6
Collateral trust	5
Atlantic & Pacific, 1st	4
Income, 1910	6
Central Division, 1st	6
Boston & Maine	7
Boston & Albany, not mortgaged	7
Burlington & Missouri River, land grant	7
In Nebraska, non-exempt, 1st	6
In Nebraska, exempt, 1st	6
In Nebraska, 1910	4
Cedar Rapids & Missouri River	7
Chicago, Burlington & Quincy, Denver Division	4
1st, 1903	7
Southwestern Division	4
Sinking fund	5
Plain bonds	4
Debentures	5
Chicago, Burlington & Northern	5
Debenture	6
Chicago, Milwaukee & St. Paul, Western Division	6
Dubuque Division	6
Chicago & Eastern Illinois, consolidated	6
Chicago, Kansas & Western	5
Incomes	
Chicago & West Michigan	5
Cincinnati, Sandusky & Cleveland, 1st	7
Columbus, Springfield & Cincinnati	7
Connecticut & Passum, 1st	7
Consolidated Railroad of Vermont	5

	Interest.
California Southern, income	6
1st, 1926	6
Detroit, Lansing & Northern, 1st	7
Dixon, Peoria & H	8
Eastern, 1906	6
Fort Scott, S. E. & M	7
Fort Scott & Gulf, 1st	7
Equipment	6
Fremont & Elk Horn	6
Iowa Falls & Sioux City	7
Illinois Grand Trunk	8
Kansas City & Camden	10
Kansas City, Topeka & Western, 1st	7
Kansas City, Em. & Southern, Gulf	7
Kansas City, Lawrence & Southern, 1st	6–5
Kansas City, Memphis & Birmingham	5
Kansas City, Clinton & Springfield	5
Kansas City, Springfield & Memphis	6
Kansas City, St. Joseph & Council Bluffs, 1st	7
Leavenworth, Topeka & Southwestern	4
Little Rock & Fort Smith, land grant, 1st	7
Maine Central, 1912	7
Marion & McPher	7
Marquette, Houghton & Ontonagon	6
1st mortgage, 1923	6
1925	6
Mexican Central, 1911	7
Income	3
Scrip	
Debenture	10
New assessment	4
Bond scrip	
New York & New England, 1st	6

	Interest.
New York & New England, 1st, 1905	7
2d mortgage	6
2d mortgage, scaled	3–5
New Mexico & Southern Pacific, 1st	7
Northern Pacific, 1st	6
P. d'O. Division	6
2d mortgage coupon	6
Ogdensburg & Lake Champlain, 1st	5
Consolidated, 1920	6
Income, 1920	3–6
Oregon Railway & Navigation Company	7
Consolidated, gold	5
Oregon Short Line	6
Portsmouth, Great Falls & Camden	4½
Pueblo & Arkansas V., 1st	7
Republican Valley	6
Rutland, 2d mortgage, equipment	5
1st mortgage	6
St. Louis, Kansas City & Southwestern	6
Sonora, 1st	7
South Kansas	5
Incomes	
Southern Kansas & Western, 1st	7
Union Pacific, 1st, gold	6
Sinking fund, 3d mortgage	8
Wisconsin Central, 2d series	7
1st series	5
Wisconsin Valley, 1st	7
Worcester, Nashua & Rochester	5

BALTIMORE STOCK-EXCHANGE.

Railroad Stocks.

	Interest.
Atlanta & Charleston	100
Baltimore & Ohio	100
1st preferred	100
2d preferred	100
Canton Co	100
Central Ohio	50
Preferred	50
Cincinnati, Washington & Baltimore	100
Preferred	100
Charlotte, Columbia & Augusta	100
Columbia & Greenville, preferred	100
Georgia Pacific	
Northern Central	50
Parkersburg Branch	50
Petersburg	100
Pittsburg & Connellsville	
Richmond, York River & Chesapeake	
Virginia Midland, 1st preferred	100
Common	100
Wilmington, Columbia & Augusta	100
Wilmington & Weldon	100
Western Maryland	50

Railroad Bonds.

Atlanta & Charleston, 1st mortgage	7
Incomes	6
Baltimore & Ohio, East Side, 1st	5
Baltimore & Ohio, extended	4
Gold	5

	Interest.
Baltimore & Potomac, 1st	6
Tunnel	6
Cape Fear & Yadkin Valley, 1st	6
Central Ohio, 1st, 1890	6
Charlotte, Columbia & Augusta, 1st	7
2d	7
Cincinnati & Baltimore	7
Cincinnati, Washington & Baltimore, 1st	4½
2d	5
3d	3
Income	5
Columbia & Greenville, 1st	6
2d, 1926	6
Georgia Pacific, 1st, 1922	6
2d, income	6
Northern Central, general	4½
Cur., 1900	6
Gold, 1900	6
Gold, 1904	6
Gold, 1926, Series A	5
Gold, 1926, Series B	5
Ohio & Mississippi, Springfield Division	7
2d	7
Consolidated sinking fund	7
General	5
Petersburg, Class A	5
Petersburg, Class B	6
Pittsburg & Con., 1st, 1898	7
Richmond & Danville, gold	6
1890	6
Piedmont Branch	8
Seaboard & Roanoke	5
Union Railroad, End. by Chattanooga Company...	

	Interest.
Virgina Midland, 1st series	6
2d series	6
3d series	5–6
4th series	3–4–5
5th series	5
Virginia and Tennessee, 4th	8
5's	5
West Virginia Central, 1st	6
Western Alabama, 1890	8
West Maryland, 3d guaranteed by city	6
Western North Carolina, 1st, 1890	7
Consolidated	6
Wilmington, Columbia & Augusta, 1910	6
Wilmington & Weldon, new	5

THE END.

SCIENTIFIC PUBLICATIONS.

Sight:

An Exposition of the Principles of Monocular and Binocular Vision. By JOSEPH LE CONTE, LL. D., author of "Elements of Geology," "Religion and Science," and Professor of Geology and Natural History in the University of California. With numerous Illustrations. 12mo. Cloth, $1.50.

"It is pleasant to find an American book which can rank with the very best of foreign works on this subject. Professor Le Conte has long been known as an original investigator in this department; all that he gives us is treated with a master-hand."—*The Nation.*

Animal Life,

As affected by the Natural Conditions of Existence. By KARL SEMPER, Professor of the University of Würzburg. With Two Maps and One Hundred and Six Woodcuts, and Index. 12mo. Cloth, $2.00.

"It appears to me that, of all the properties of the animal organism, Variability is that which may first and most easily be traced by exact investigation to its efficient causes; and as it is beyond a doubt the subject around which, at the present moment, the strife of opinions is most violent, it is that which will be most likely to repay the trouble of closer research. I have endeavored to facilitate this task so far as in me lies."—*From the Preface.*

The Atomic Theory.

By AD. WURTZ, Membre de l'Institut; Doyen Honoraire de la Faculté de Médecine; Professeur à la Faculté des Sciences de Paris. Translated by E. CLEMINSHAW, M. A., F. C. S., F. I. C., Assistant Master at Sherborne School. 12mo. Cloth, $1.50.

"There was need for a book like this, which discusses the atomic theory both in its historic evolution and in its present form. And perhaps no man of this age could have been selected so able to perform the task in a masterly way as the illustrious French chemist, Adolph Wurtz. It is impossible to convey to the reader, in a notice like this, any adequate idea of the scope, lucid instructiveness, and scientific interest of Professor Wurtz's book. The modern problems of chemistry, which are commonly so obscure from imperfect exposition, are here made wonderfully clear and attractive."—*The Popular Science Monthly.*

Education as a Science.

By ALEXANDER BAIN, LL. D. 12mo. Cloth, $1.75.

"This work must be pronounced the most remarkable discussion of educational problems which has been published in our day. It should be in the hands of every school-teacher and friend of education throughout the land."—*New York Sun.*

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

SCIENTIFIC PUBLICATIONS.

Suicide:

An Essay in Comparative Moral Statistics. By HENRY MORSELLI, Professor of Psychological Medicine in the Royal University, Turin. 12mo. Cloth, $1.75.

"A most valuable contribution to English literature touching a theme most distressing in the act and terrible in its consequences, yet to this hour but very imperfectly studied or understood."—*Philadelphia Times.*

Volcanoes:

What they Are and what they Teach. By J. W. JUDD, Professor of Geology in the Royal School of Mines (London). With Ninety-six Illustrations. 12mo. Cloth, $2.00.

"In no field has modern research been more fruitful than in that of which Professor Judd gives a popular account in the present volume. The great lines of dynamical, geological, and meteorological inquiry converge upon the grand problem of the interior constitution of the earth, and the vast influence of subterranean agencies. . . . His book is very far from being a mere dry description of volcanoes and their eruptions; it is rather a presentation of the terrestrial facts and laws with which volcanic phenomena are associated."—*Popular Science Monthly.*

The Sun:

By C. A. YOUNG, Ph. D., LL. D., Professor of Astronomy in the College of New Jersey. With numerous Illustrations. Third edition, revised, with Supplementary Note. 12mo. Cloth, $2.00.

The "Supplementary Note" gives important developments in solar astronomy since the publication of the second edition in 1882.

"There is no rhetoric in his book; he trusts the grandeur of his theme to kindle interest and impress the feelings. His statements are plain, direct, clear, and condensed, though ample enough for his purpose, and the substance of what is generally wanted will be found accurately given in his pages."—*Popular Science Monthly.*

Illusions:

A Psychological Study. By JAMES SULLY, author of "Sensation and Intuition," etc. 12mo. Cloth, $1.50.

"An interesting contribution by Mr. James Sully to the study of mental pathology. The author's field of inquiry covers all the phenomena of illusion observed in sense-perception, in the introspection of the mind's own feelings, in the reading of others' feelings, in memory, and in belief. The author's conclusions are often illustrated by concrete example or anecdote, and his general treatment of the subject, while essentially scientific, is sufficiently clear and animated to attract the general reader."—*New York Sun.*

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

SCIENTIFIC PUBLICATIONS.

The Brain and its Functions.

By J. LUYS, Physician to the Hospice de la Salpêtrière. With Illustrations. 12mo. Cloth, $1.50.

"No living physiologist is better entitled to speak with authority upon the structure and functions of the brain than Dr. Luys. His studies on the anatomy of the nervous system are acknowledged to be the fullest and most systematic ever undertaken. Dr. Luys supports his conclusions not only by his own anatomical researches, but also by many functional observations of various other physiologists, including of course Professor Ferrier's now classical experiments."—*St. James's Gazette.*

The Concepts and Theories of Modern Physics.

By J. B. STALLO. 12mo. Cloth, $1.75.

"Judge Stallo's work is an inquiry into the validity of those mechanical conceptions of the universe which are now held as fundamental in physical science. He takes up the leading modern doctrines which are based upon this mechanical conception, such as the atomic constitution of matter, the kinetic theory of gases, the conservation of energy, the nebular hypothesis, and other views, to find how much stands upon solid empirical ground, and how much rests upon metaphysical speculation. Since the appearance of Dr. Draper's 'Religion and Science,' no book has been published in the country calculated to make so deep an impression on thoughtful and educated readers as this volume. . . . The range and minuteness of the author's learning, the acuteness of his reasoning, and the singular precision and clearness of his style, are qualities which very seldom have been jointly exhibited in a scientific treatise."—*New York Sun.*

The Formation of Vegetable Mould,

Through the Action of Worms, with Observations on their Habits. By CHARLES DARWIN, LL. D., F. R. S., author of "On the Origin of Species," etc., etc. With Illustrations. 12mo. Cloth, $1.50.

"Mr. Darwin's little volume on the habits and instincts of earthworms is no less marked than the earlier or more elaborate efforts of his genius by freshness of observation, unfailing power of interpreting and correlating facts, and logical vigor in generalizing upon them. The main purpose of the work is to point out the share which worms have taken in the formation of the layer of vegetable mould which covers the whole surface of the land in every moderately humid country. All lovers of nature will unite in thanking Mr. Darwin for the new and interesting light he has thrown upon a subject so long overlooked, yet so full of interest and instruction, as the structure and the labors of the earth-worm."—*Saturday Review.*

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

SCIENTIFIC PUBLICATIONS.

Ants, Bees, and Wasps.

A Record of Observations on the Habits of the Social Hymenoptera. By Sir JOHN LUBBOCK, Bart., M. P., F. R. S., etc., author of "Origin of Civilization, and the Primitive Condition of Man," etc., etc. With Colored Plates. 12mo, cloth, $2.00.

"This volume contains the record of various experiments made with ants, bees, and wasps during the last ten years, with a view to test their mental condition and powers of sense. The principal point in which Sir John's mode of experiment differs from those of Huber, Forel, McCook, and others, is that he has carefully watched and marked particular insects, and has had their nests under observation for long periods—one of his ants' nests having been under constant inspection ever since 1874. His observations are made principally upon ants because they show more power and flexibility of mind; and the value of his studies is that they belong to the department of original research."

Diseases of Memory.

An Essay in the Positive Psychology. By TH. RIBOT, author of "Heredity," etc. Translated from the French by WILLIAM HUNTINGTON SMITH. 12mo, cloth, $1.50.

"M. Ribot reduces diseases of memory to law, and his treatise is of extraordinary interest."—*Philadelphia Press.*

"Not merely to scientific, but to all thinking men, this volume will prove intensely interesting."—*New York Observer.*

"M. Ribot has bestowed the most painstaking attention upon his theme, and numerous examples of the conditions considered greatly increase the value and interest of the volume."—*Philadelphia North American.*

Myth and Science.

By TITO VIGNOLI. 12mo, cloth, price, $1.50.

"His book is ingenious; . . . his theory of how science gradually differentiated from and conquered myth is extremely well wrought out, and is probably in essentials correct."—*Saturday Review.*

"The book is a strong one, and far more interesting to the general reader than its title would indicate. The learning, the acuteness, the strong reasoning power, and the scientific spirit of the author, command admiration."—*New York Christian Advocate.*

"An attempt made, with much ability and no small measure of success, to trace the origin and development of the myth. The author has pursued his inquiry with much patience and ingenuity, and has produced a very readable and luminous treatise."—*Philadelphia North American.*

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

THE

POPULAR SCIENCE MONTHLY.

ESTABLISHED BY EDWARD L. YOUMANS.

Edited by W. J. YOUMANS.

THE POPULAR SCIENCE MONTHLY was established a dozen years ago to bring before the general public the results of scientific thought on many large and important questions which could find no expression in the current periodicals. Scientific inquiry was penetrating many new fields, extending important knowledge, and profoundly affecting opinion upon numberless questions of speculative and practical interest. It was the policy of this magazine at the outset, and has been constantly adhered to since, to obtain the ablest statements from the most distinguished scientific men of all countries in their bearing upon the higher problems of investigation. Leaving the dry and technical details of science, which are of chief concern to specialists, to the journals devoted to them, THE POPULAR SCIENCE MONTHLY has dealt with those more general and practical subjects which are of the greatest interest and importance to the people at large.

Science is the great agency of improvement in this age, private and public, individual, social, professional, and industrial. In its irresistible progress it touches everywhere, and affects everybody. It gives law to the material interests of the community, and modifies its ideas, opinions, and beliefs, so that all have an interest in being informed of its advancement. Those, therefore, who desire to know what is going on in the world of thought in these stirring times, when new knowledge is rapidly extending, and old errors are giving way, will find that they can keep informed only by subscribing for THE POPULAR SCIENCE MONTHLY.

TERMS: $5 00 per annum, or 50 cents a number; cloth covers for the volumes, 50 cents each.

The volumes begin with May and November of each year, and subscriptions may begin at any time.

A club of five will be supplied for $20.00 per annum.

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

THE

NEW YORK MEDICAL JOURNAL,

A Weekly Review of Medicine.

BEGINNING with the number for Saturday, January 6, 1883, the "New York Medical Journal" became a weekly instead of a monthly. The frequent publication of LECTURES is now made a prominent feature, and pains are taken to choose such as will prove valuable to the reader. At the same time care is taken not to lessen the variety and value of the more formal articles known as ORIGINAL COMMUNICATIONS. In accepting articles of that class, regard is had more particularly to the wants of the general practitioner, and all the special branches of medicine are duly represented.

In the matter of BOOK NOTICES, publishing more frequently, we are able to avoid delays. Formerly a great deal of contemporary literature was kindly sent to us in the reasonable expectation that it would be noticed in our pages—and yet we were obliged to pass it by. Much of this matter is of such a nature that, if noticed at all, it should be noticed without delay. The columns of the weekly now enable us to do justice to the literature of the day, and to keep our readers fully and promptly informed of its scope and character.

CLINICAL REPORTS will hereafter be a regular feature of the weekly Journal, embracing clinical records from the various hospitals and clinics, not only of New York, but of various other cities, together with clinical contributions from private practice.

The EDITORIAL ARTICLES will be more numerous, and we shall be able to give more timely consideration to passing events. It is in this department especially that frequency of issue adds to the interest of a journal.

SOCIETY PROCEEDINGS will be given more promptly than before, and those of a greater number of societies will figure.

The REPORTS ON THE PROGRESS OF MEDICINE—a feature of the Journal which we have reason to think has been highly valued by our readers—will not be curtailed in their scope. On the contrary, an additional quarterly report will be published on *Preventive Medicine and Medical Jurisprudence.* All the reports will be given quarterly, so that upon each of the departments of medicine formerly covered by them (anatomy and physiology; materia medica, therapeutics, and toxicology; general medicine; surgery; orthopædic surgery and diseases of the joints; obstetrics and gynæcology; ophthalmology and otology; cutaneous and venereal diseases; and psychological medicine), as well as the new topic mentioned above, a report will appear four times a year.

Under the head of MISCELLANY we shall include matters of news, correspondence, etc.—the varied items that one expects to find in a weekly paper.

Terms, $5.00 per annum (postage prepaid by the Publishers); single copies, 12 cents.

New York: D. APPLETON & CO., 1, 3, & 5 Bond Street.

THIS BOOK IS DUE ON THE LAST DATE
STAMPED BELOW

AN INITIAL FINE OF 25 CENTS
WILL BE ASSESSED FOR FAILURE TO RETURN
THIS BOOK ON THE DATE DUE. THE PENALTY
WILL INCREASE TO 50 CENTS ON THE FOURTH
DAY AND TO $1.00 ON THE SEVENTH DAY
OVERDUE.

OCT 11 1934

DEC 20 1937